

Staging as a Monitoring Device under Endogenous Deadlines*

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Abstract

We study a principal-agent problem where the principal dynamically chooses between two methods for solving a problem: a direct execution route and a multi-stage training route with an observable milestone. To mitigate moral hazard, the principal commits to an endogenously determined deadline. The optimal contract is shaped by the interplay of three forces: the milestone effect from the training route’s monitoring advantage, the deadline effect that favors the simpler execution route as time runs short, and the relative efficiency of each path.

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1 Introduction

In managing complex projects, a central challenge is choosing the best methodology. A manager must often decide between pursuing a direct, single-stage approach using existing capabilities and adopting a more complex, multi-stage strategy that first requires developing a specialized skill. This tradeoff recurs in many settings. A software company might rely on its existing codebase to build a product, or it could first train its team on a new programming language tailored to the project’s requirements. Similarly, a graduate student writing a dissertation might draw only on the training from their first- and second-year courses, or they might first build new competencies through additional coursework outside their department.

Perhaps most consequentially, this tradeoff was at the heart of the race to develop COVID-19 vaccines. Some ventures—Oxford-AstraZeneca, Johnson & Johnson, and Novavax—pursued established approaches such as viral-vector and protein-subunit methods. By contrast, Pfizer-BioNTech and Moderna-NIAID took the novel route of developing an mRNA platform, a strategy that required mastering the new technology before an actual vaccine could be produced.

To study this type of trade-off, we develop a model where a principal hires an agent to solve a problem. At any moment, the principal instructs the agent to pursue one of two methodologies. She can recommend the **execution route**, a direct path where the agent applies a basic skill to solve the problem, with success modeled as a Poisson arrival. Alternatively, she can recommend the **training route**, an indirect, two-stage path. On this route, the agent must first acquire an advanced skill before using it to solve the problem; both the initial skill acquisition and the final solution are also governed by Poisson processes.

A crucial feature of the training route is that the acquisition of the advanced skill is a *publicly observable* milestone, providing a clear signal of progress. This assumption is grounded in practice, where such preparatory stages often conclude with verifiable outcomes, such as obtaining a professional certification, completing a required training program, or a successful prototype demonstration.

The choice of methodology is complicated by a classic moral hazard problem. The solution or skill acquisition only occurs if the agent exerts effort, which is his private information. If the agent shirks, he enjoys a private benefit, and the principal’s ability to counteract this is limited by the agent’s limited liability, which restricts her to offering non-negative rewards. These frictions make deadlines essential for providing incentives; without one, the agent could shirk indefinitely. Therefore, the principal must impose a deadline, which is *endogenously* determined as part of the optimal contract.

The principal’s optimal strategy is governed by the interplay of three competing economic forces. The first is the **milestone effect**, where the observable progress in the training route provides a monitoring advantage that allows for deadline extensions. This is weighed against the **deadline effect**, which makes the single-stage execution route more appealing as time runs short, since a single breakthrough is more probable than the two breakthroughs required for the training route. The final force is **efficiency**, defined by the expected time required to solve the problem. To isolate the core tensions, we first analyze a benchmark where both routes are equally efficient (i.e., have the same expected completion time) and later introduce the possibility that the execution route is more efficient (i.e., has a shorter expected completion time), creating a further trade-off against the training route’s monitoring benefits.

Focusing on the case where the execution and training routes are equally efficient, we find that the optimal contract depends critically on the project return—the gross value of the solution relative to the operating cost (Theorem 1). When the project return is low, the optimal deadline is short, and the principal opts exclusively for the execution path. Conversely, when the project return is high, the benefit of monitoring the agent’s progress becomes crucial, causing the milestone effect to dominate the deadline effect; thus, the principal always recommends the training route. For projects with intermediate returns, the optimal contract involves a switch in strategy: the principal initially recommends the training path but instructs the agent to switch to the execution path if the advanced skill is not acquired and the deadline becomes imminent.

We then introduce an efficiency loss from training, reflecting scenarios where its multi-stage nature may be slower than the more direct execution route. When this efficiency loss is small, our previous characterization remains robust; the optimal contract is still determined by low, intermediate, or high values of the project return (Proposition 4). However, when the efficiency loss is large, the trade-offs become sharper. The execution route’s efficiency advantage makes it the preferred choice not only near the deadline but also at the beginning of a long project. Consequently, for most project returns, an execution-only contract is optimal. More interestingly, for very high-return projects, a novel two-switch contract emerges where all three economic forces shape the outcome: the principal begins with the execution route (due to efficiency), switches to the training route in the middle of the contract to leverage the milestone effect for monitoring, and finally reverts to the execution route as the deadline looms (Proposition 5).

Our findings align with observed differences in the structure of applied and basic scientific research. Applied research, such as developing a new drug or conducting clinical trials, is typically structured in stages with clear milestones. This corresponds to the training route in our model, which we find is optimal for high-return projects—a fitting description for applied research where the potential payoff is large relative to the operating cost. In contrast, basic research is often pursued “without thought of practical ends” (Bush, 1945), and thus has a lower immediate project return.¹ This prediction is consistent with funding mechanisms like the National Institutes of Health’s R01 grant, which supports “a discrete, specified, circumscribed project” rather than a staged one.²

Related Literature There is a growing literature on contracting for multi-stage projects, e.g., Hu (2014); Green and Taylor (2016a); Wolf (2018); Moroni (2022). The most closely

¹Bush argues that although broad and basic studies seem to be less important than applied ones, they are essential to combat diseases because progress in the treatment “will be made as the result of fundamental discoveries in subjects unrelated to those diseases, and perhaps entirely unexpected by the investigator.” However, since this article does not consider externalities, we abstract from this possibility and focus on the principal’s return from the completed project.

²<https://grants.nih.gov/grants/funding/r01.htm>

related study is [Green and Taylor \(2016a\)](#), who study a model in which multiple breakthroughs are needed to complete a project and in which an agent must be incentivized to exert unobservable effort. The training route considered here comprises the baseline model with the tangible breakthrough in the working paper version of their paper ([Green and Taylor, 2016b](#)). However, the option to complete the project directly, which is not considered in their setup, allows the principal to face a choice problem between the two routes. Moreover, this choice problem arises at every point in time. Therefore, the principal’s problem is more complex from a dynamic perspective.

A related article is [Carnehl and Schneider \(2023\)](#), where they explore a two-armed bandit problem with one arm requiring a single breakthrough and the other needing multiple breakthroughs. The agent knows the arrival rates for the latter but must infer the feasibility of the former through experimentation—a key difference between their paper and this one. Unlike their focus on a single-agent decision problem with an exogenous deadline, this paper addresses principal-agent contracting with an endogenously determined deadline. Despite these differences, we share a common insight in that the chosen approaches may switch up to two times.

This investigation is also related to the literature on monitoring in dynamic contracts, e.g., [Orlov \(2022\)](#); [Piskorski and Westerfield \(2016\)](#); [Dilmé and Garrett \(2019\)](#); [Marinovic and Szydlowski \(2022\)](#); [Varas et al. \(2020\)](#); [Marinovic and Szydlowski \(2023\)](#); [Chen et al. \(2020\)](#); [Wong \(2023\)](#). In most of these papers, a monitoring process provides some information on the agent’s current or past action. In this sense, the completion of the first breakthrough (advanced skill acquisition) in the training route can be considered as a monitoring device since it lets the principal know that the agent has worked. However, the skill acquisition gives more information than merely the agent’s past actions. Before the skill acquisition, the success requires one relatively hard breakthrough or two easier breakthroughs. After skill acquisition, it requires only one relatively easy breakthrough. Thus, the skill acquisition is distinguished from standard monitoring processes since it also provides information about

the subsequent likelihood of success.

This work is also relevant to the literature studying complementary innovations, e.g., [Green and Scotchmer \(1995\)](#); [Gilbert and Katz \(2011\)](#); [Bryan and Lemus \(2017\)](#); [Poggi \(2021\)](#). Two subprojects in the sequential approach can be considered as ‘perfect’ complements in the sense that completing one task in the training route does not create any value but completing both of them does. The most relevant paper in this line is [Kim and Poggi \(2025\)](#), which introduces an innovation race model with two R&D routes: one requiring a single breakthrough (direct development) and the other requiring two breakthroughs (research and development). However, to our knowledge, most studies in this literature focus on the problems involving competing firms or a single decision maker, whereas this article addresses an agency problem.

2 Setup

Environment. A principal (she) hires an agent (he) to complete a project. Work takes place in continuous time over an unbounded horizon, $t \in [0, \infty)$. Completing the project yields the principal a gross payoff $\Pi > 0$ and ends the game; until the project is completed or abandoned she incurs an operating flow cost $c > 0$. Both players are risk neutral, do not discount the future, and have outside options normalized to zero. The principal has unlimited resources with which to fund the project; the agent is protected by limited liability, so every transfer to him must be nonnegative.

Effort. The agent’s action at time t is his effort $\beta_t \in [0, 1]$, where $\beta_t = 1$ is full effort and $\beta_t = 0$ is full shirking.³ Effort is his private information. Withholding the fraction $1 - \beta_t$ leaves him a private flow benefit $(1 - \beta_t)\phi$, and we assume $c > \phi > 0$, so that the cost of

³Both the arrival rate and the private benefit are linear in β_t , so the agent’s payoff is linear in his action and a corner is optimal. It is therefore without loss to restrict attention to $\beta_t \in \{0, 1\}$. ** YG: TBA: See [Appendix ?? for details.](#)**

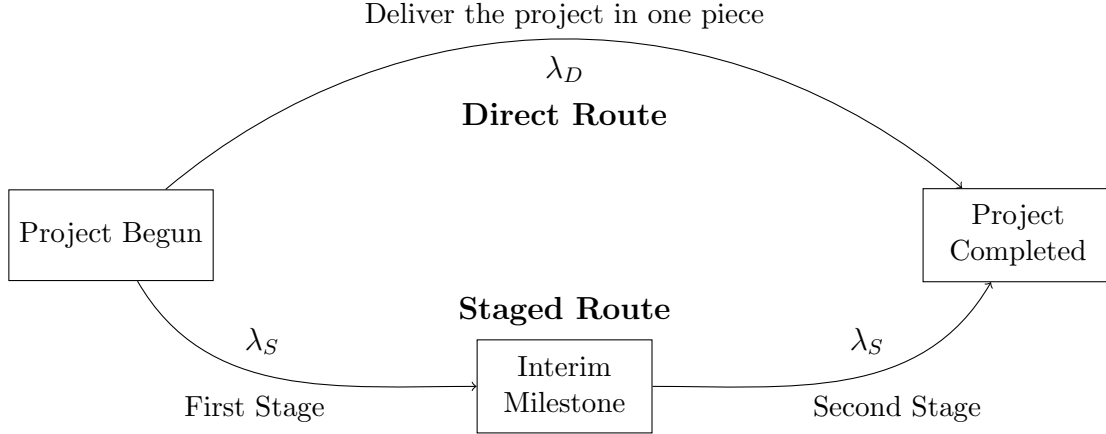


Figure 1: Routes to Project Completion

keeping the project alive exceeds the agent's gain from shirking.⁴

Routes. At each instant the principal directs the agent along one of two routes to completion. Under the **direct route**, the agent works on the project as a whole and delivers it in finished form; completion arrives at Poisson rate $\lambda_D \beta_t$. Under the **staged route**, he first produces an *interim milestone*—a prototype, a demonstration, a report on work in progress—and only then completes the project; each stage arriving at rate $\lambda_S \beta_t$. Figure 1 illustrates the two paths. We assume $\lambda_S > \lambda_D$: otherwise, the staged route would be slower stage for stage as well as in total, and would never be used.

Staging is not a technological necessity: the same project can be finished either way, and the interim milestone exists only because the contract calls for it. What it does is generate a verifiable intermediate output along the way. The milestone and the completed project are publicly observable and verifiable; the agent's effort is not. Payments may therefore be conditioned on which deliveries have occurred and when, and, absent any delivery, calendar time is the only payoff-relevant public information.

⁴The friction is familiar from models of cash diversion, e.g. DeMarzo and Sannikov (2006): the principal outlays c per unit of time, and during an instant of full shirking the agent privately captures ϕ of that outlay. In that reading ϕ/c is the diversion rate and $1 - \phi/c$ the resulting deadweight wedge.

Efficiency. Absent any agency friction, and with no deadline in force, the choice between routes would turn on expected completion time alone. The direct route completes the project in expected time $1/\lambda_D$, at expected cost c/λ_D ; the staged route requires two successes and so takes $2/\lambda_S$, at expected cost $2c/\lambda_S$. The first best is therefore $\Pi - \min\{c/\lambda_D, 2c/\lambda_S\}$.

Sections 3 and 4 focus on the case in which the two routes are equally efficient, $\lambda_S = 2\lambda_D$. Staging is then *technologically innocuous*: a principal who could dictate effort would be indifferent between the routes, so any staging that appears in the optimal contract is there for incentive reasons alone. Section 5 relaxes this assumption, allowing staging to be costly: $\lambda_S < 2\lambda_D$.⁵

We also assume the project is worth undertaking:

$$\Pi > \frac{c}{\lambda_D} \quad \Longleftrightarrow \quad \pi \equiv \frac{\lambda_D \Pi}{c} > 1, \quad (2.1)$$

where π denotes the *project return*.

Deadlines. Limited liability makes deadlines indispensable. Were the contract never to terminate, the agent could shirk at every date and still be retained, collecting ϕ per unit of time in perpetuity; because payments cannot be negative, no finite reward for completion would deter him.⁶ Every incentive-compatible contract must therefore abandon the project in finite time, trading the surplus lost when a viable project is abandoned against the rents the principal must otherwise concede.

Contracts. At $t = 0$ the principal offers a contract and commits fully to its terms; if the agent rejects, both parties receive zero. We restrict attention to contracts that recommend full effort at every date, which is without loss of generality: because $c > \phi$, any interval on

⁵If $\lambda_S > 2\lambda_D$, staging would be the more efficient route. We exclude that case because our interest is in the tension between the two routes: if the direct route drops out of the optimal contract entirely, the problem reduces to the single-route analysis of Green and Taylor (2016b).

⁶Absent limited liability the principal could implement the first best by selling the project to the agent upfront at a price equal to its first-best value derived above, and leaving him the residual. Limited liability implies that the agent lacks the resources to make such a payment.

which the agent is meant to shirk can be excised, advancing all subsequent events and saving $c - \phi$ per unit of time.

A (deterministic) contract is an object $\Gamma \equiv \{T, \{a_t, R_t, \hat{\Gamma}^t\}_{0 \leq t \leq T}\}$, whose components are defined at calendar time t as follows.⁷

1. $T \in \mathbb{R}_+$: the deadline at which the project is abandoned if no delivery has occurred.
2. $a_t \in \{0, 1\}$: the route recommended at t , where $a_t = 1$ denotes the *direct* route and $a_t = 0$ the *staged* route.
3. $R_t \geq 0$: the payment to the agent if he completes the project at t without having delivered an interim milestone.⁸
4. $\hat{\Gamma}^t \equiv \{T^t, \{R_s^t\}_{t \leq s \leq T^t}\}$: the continuation contract taking effect if the milestone arrives at t , consisting of a revised deadline $T^t \geq t$ and payments $R_s^t \geq 0$ for completing the project at s .

Any delivery inconsistent with the recommended route—a milestone under the direct route, or a completed project with no milestone on record under the staged route—terminates the contract without payment, so the agent follows the recommendation. Note that T^t is unrestricted: that the principal extends the deadline, and by how much, is a result rather than an assumption.

3 Observable Effort with Exogenous Deadlines

Before turning to the contracting problem under agency, we examine how the principal deploys the two routes when the agent's effort is observable and contractible, focusing on the benchmark case where staging is technologically innocuous ($\lambda_S = 2\lambda_D$). Because effort can be directly mandated, no obedience (incentive) constraint arises, and route choice is

⁷Randomization cannot improve upon this class, though the argument requires care; see Remark 1.

⁸Since both parties are risk neutral and do not discount, all payments may be backloaded to the moment of a delivery without loss of generality (see, e.g., Ray, 2002). Nonnegativity is limited liability.

governed purely by the technological trade-offs between the two paths. Absent a deadline, observable and contractible effort delivers the first-best value, and under equal efficiency, the principal is indifferent between the two routes.

An interesting trade-off arises in the presence of an exogenous deadline T . Suppose further that the deadline is extended by $\Delta \geq 0$ upon delivery of the interim milestone prior to T .⁹ The principal thus faces two effective deadlines: T if no milestone has been delivered, and $T + \Delta$ if one has. These deadlines introduce inefficiency: whenever the project is worth undertaking (i.e., when (2.1) holds), efficiency demands running it until completion without truncation.

Given (T, Δ) , the principal chooses a route policy $a : [0, T] \rightarrow \{0, 1\}$ specifying which route to use at each calendar date. Her goal is to select an optimal policy that maximizes her expected payoff. This analysis provides key intuition for the optimal contract under moral hazard and is also of independent interest given the ubiquity of deadlines in project management.

3.1 Deadline and Milestone Effects

To build intuition for route choice, this subsection elaborates two fundamental forces that govern the principal's trade-off under deadlines: the *deadline effect* and the *milestone effect*. To illustrate these mechanisms, we compare two benchmark policies that never switch: the *direct-only* policy ($a_t = 1$ for all $t \in [0, T]$), under which the agent works on the project as a whole throughout, and the *staged-only* policy ($a_t = 0$ for all $t \in [0, T]$), under which he works toward the milestone and then, if delivered, toward project completion. While the principal's expected payoff also reflects operating costs incurred prior to completion or abandonment—which depend on the chosen policy—the probability of completing the project in time plays a central, decisive role in shaping her preference.

⁹The case $\Delta = 0$ is permitted, so that staging confers no additional runway. One could also allow $\Delta < 0$; we show in Section 4 that under agency it is optimal to extend, not curtail, the deadline upon delivery of the milestone.

Under the direct-only policy, a single success is needed by date T , yielding the completion probability

$$\mathcal{P}^D(T) \equiv \int_0^T \lambda_D e^{-\lambda_D \tau} d\tau = 1 - e^{-\lambda_D T}. \quad (3.1)$$

Under the staged-only policy, the milestone must arrive at some date $\tau_m \leq T$, and the project must then be completed by the extended deadline $T + \Delta$, yielding the completion probability

$$\mathcal{P}^S(T, \Delta) \equiv \int_0^T \left[\int_{\tau_m}^{T+\Delta} \lambda_S e^{-\lambda_S(\tau-\tau_m)} d\tau \right] \lambda_S e^{-\lambda_S \tau_m} d\tau_m = 1 - (1 + \lambda_S T e^{-\lambda_S \Delta}) e^{-\lambda_S T}. \quad (3.2)$$

Each route possesses a distinct advantage. Setting $\Delta = 0$ isolates the technological trade-off by stripping the staged route of any deadline extension.

Lemma 1 (Deadline effect). *Suppose $\Delta = 0$. There exists $\check{T} > 0$ such that $\mathcal{P}^D(T) > \mathcal{P}^S(T, 0)$ for all $T < \check{T}$, and $\mathcal{P}^S(T, 0) > \mathcal{P}^D(T)$ for all $T > \check{T}$.*

The direct route's advantage lies in requiring only a single arrival. Over a short horizon, the probability of one arrival is first-order in remaining time, whereas the probability of two arrivals is second-order; when time is short, needing one slow success dominates needing two fast ones. Conversely, the staged route's advantage is that its two individual stages are faster. Over a long horizon, achieving two fast successes is more likely than achieving one slow one.

Panel (a) of Figure 2 illustrates the completion-time densities for both routes. Under equal efficiency ($\lambda_S = 2\lambda_D$), both densities share the same mean (marked by the dotted line). However, the direct route places more mass both near zero and in the far tail (shaded area), whereas the staged route concentrates completion tightly around the mean. Lemma 1 formalizes the observation that the direct route's completion time is a mean-preserving spread of the staged route's: a tight deadline penalizes dispersion (favoring the direct route), whereas a generous deadline rewards concentration (favoring the staged route). We refer to this force as the *deadline effect*.

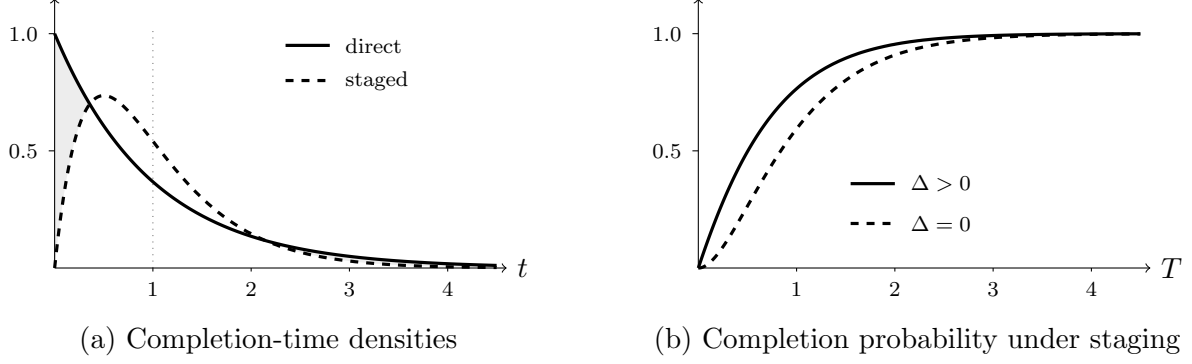


Figure 2: Completion times under equal efficiency ($\lambda_D = 1$, $\lambda_S = 2$)

The deadline extension Δ introduces a second, distinct advantage for the staged route. Because $\mathcal{P}^S(T, \Delta)$ is strictly increasing in Δ while $\mathcal{P}^D(T)$ is independent of it, any $\Delta > 0$ shifts the payoff balance in favor of staging. Panel (b) plots the staged completion probability with and without an extension. This advantage stems not from technology, but from information: the milestone reveals to the principal that concrete progress has occurred, and the extension Δ represents the value of that information. We refer to this force as the *milestone effect*.

3.2 Optimality of One-Switch Policy

Building on the intuition of the deadline and milestone effects developed in Section 3.1, we now introduce a class of policies that dynamically leverages the respective advantages of both routes. The previous discussion suggests that the staged route is highly effective when the deadline is distant (capitalizing on the milestone effect), whereas the direct route is preferable when the deadline is nigh (capitalizing on the deadline effect). This naturally motivates a policy that begins with staging and falls back to the direct route if time begins to run short.

Definition 1. A policy is called a *one-switch policy* if there exists an intermediate deadline $S \in [0, T]$ such that: (i) the principal chooses the staged route up to S ($a_t = 0$ for $t \leq S$); (ii) if the interim milestone is delivered before S , she proceeds with the second stage until

the extended deadline $T + \Delta$; and (iii) if the milestone is not delivered by S , she switches to the direct route until the original deadline T .

Note that the benchmark policies of the previous section also takes a form of one-switch policy: the direct-only policy sets $S = 0$ and the staged-only policy sets $S = T$.

The following proposition establishes that the optimal policy invariably takes the form of a one-switch policy. We prove this result utilizing a guess-and-verify approach to solve the principal's dynamic optimization problem.

Proposition 1. *Suppose that the two routes are equally efficient ($\lambda_S = 2\lambda_D$). When the principal faces a deadline T and a deadline extension Δ contingent on milestone delivery, the optimal policy is characterized as follows:*

- (a) **(Long extension)** *If $\Delta \geq \bar{\Delta} \equiv \frac{1}{\lambda_S} \log \left[\frac{2\pi-1}{\pi-1} \right]$, the staged-only policy is optimal.*
- (b) **(Short extension)** *If $\Delta < \bar{\Delta}$, there exists a cutoff $\hat{T} > 0$ such that:*
 - (i) *when $T < \hat{T}$, the direct-only policy is optimal;*
 - (ii) *when $T > \hat{T}$, the one-switch policy with the intermediate deadline $S = T - \hat{T}$ is optimal.*

This proposition implies that the direct route is never used when the deadline extension is sufficiently generous. On the other hand, when the deadline extension is relatively short, there exists a critical time threshold $\hat{T}$ such that the principal switches to the direct route when fewer than $\hat{T}$ units of time remain before the deadline.

Figure 3 illustrates the relative advantages of the two benchmark policies. Based on the deadline and milestone effects, we can infer that the direct-only policy yields a higher completion probability than the staged-only policy *only* when both the original deadline and the deadline extension are short.

With a long deadline extension (Panel 3a), the milestone effect outweighs the deadline effect globally, meaning the staged-only policy dominates in terms of completion probability

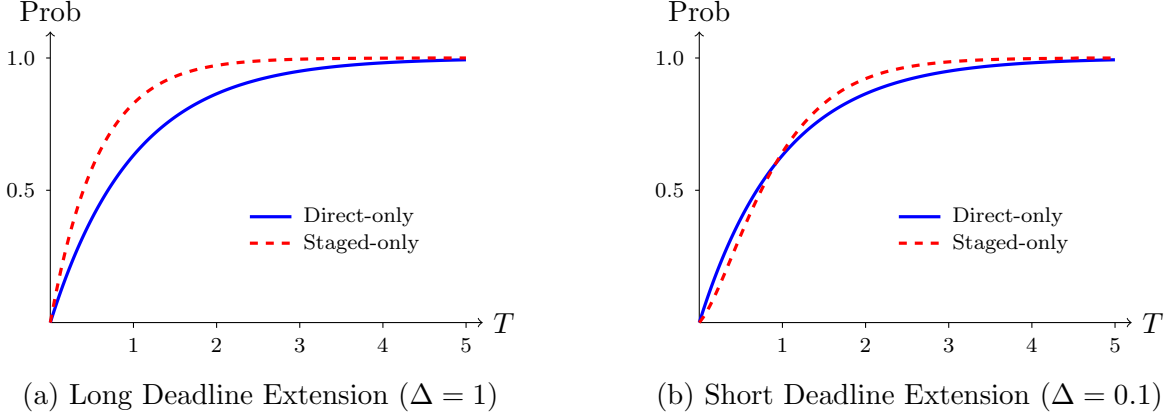


Figure 3: Completion Probabilities ($\lambda_S = 2$, $\lambda_D = 1$, $\Pi = 3$, $c = 1$)

for all horizons T . Conversely, when the extension is short (Panel 3b), the direct-only policy provides a higher completion probability for short deadlines (due to the deadline effect), whereas the staged-only policy overtakes it for long deadlines.

These competing effects play a crucial role in comparing the expected surpluses between the two policies. Because the direct-only policy can achieve a higher expected surplus only when both the deadline and the extension are short, it naturally follows that the principal's optimal strategy is a one-switch policy: utilizing the staged route while time is plentiful, and pivoting to the direct route only as the deadline tightly constrains the remaining window. Proposition 1 formally confirms that this is indeed the case.

Comparative statics of the extension cutoff Proposition 1 establishes a critical cutoff for the deadline extension, $\bar{\Delta}$, such that the staged-only policy is optimal whenever the extension exceeds this threshold; otherwise, the direct route is employed as the deadline draws near.

Observe that taking the derivative of this cutoff with respect to the project return yields

$$\frac{\partial \bar{\Delta}}{\partial \pi} = \frac{1}{\lambda_S} \left[\frac{2}{2\pi - 1} - \frac{1}{\pi - 1} \right] = -\frac{1}{\lambda_S(2\pi - 1)(\pi - 1)} < 0,$$

which is strictly negative since $\pi > 1$ by assumption (2.1). This establishes the following

lemma.

Lemma 2. *Suppose that $\lambda_S = 2\lambda_D$ and $\pi > 1$. The extension cutoff $\bar{\Delta}$ is strictly decreasing in the project return π .*

The economic intuition for this result is straightforward: as the project return π increases, maximizing the completion probability becomes increasingly critical to the principal’s expected surplus. This strongly favors the staged route, which guarantees a higher completion probability over a long horizon (due to the milestone effect). Consequently, the minimum deadline extension $\bar{\Delta}$ required to make the staged-only policy globally optimal strictly decreases as the project becomes more lucrative.

4 Agency

4.1 The Optimal Contract

In this subsection, we characterize the optimal contract in the case where the routes are equally efficient. As in the tangible progress case in the mimeo [Green and Taylor \(2016b\)](#), the optimal contract can be implemented with three key properties: (i) the contract is terminated after a deadline; (ii) the reward for the project completion, R_t , linearly diminishes over time; and (iii) the deadline is extended by $1/\lambda_S$ upon advanced skill acquisition.¹⁰

Since the contract involves a deadline and an extension upon skill acquisition, Theorem 1 suggests that the optimal choice of approaches over time likely either involve one switch from the training route to the execution route—or no switch at all. In light of this conjecture, we define contracts involving the above characteristics as follows.

Definition 2. A contract is called a *one-switch contract* with a final deadline T and an intermediate deadline $S \in (0, T)$ if

¹⁰The details of these properties will be addressed in Section 4.3. What distinguishes this work from [Green and Taylor \(2016b\)](#) is the presence of the two routes for solving the problem and the principal’s choice of which to recommend at each moment in time.

- (i) the agent is recommended to acquire the advanced skill (training route) by S ,
- (ii) when the advanced skill is acquired before S , the contract is extended by $1/\lambda_S$ and the reward upon solution at time t is $R_t^A \equiv \phi(T - t + 2/\lambda_S)$,
- (iii) if the advanced skill is not acquired by S , the agent is recommended to switch and solve the problem using the basic skill (execution route) by T and the reward upon solution at time t is $R_t^B \equiv \phi(T - t + 1/\lambda_D)$, and
- (iv) the contract is terminated if the problem is not solved by the deadlines ($T + 1/\lambda_S$ for (ii) and T for (iii)).

When $S = T$, we call the contract a *training-only contract*, and when $S = 0$, we call the contract a *execution-only contract*.

The following theorem shows that the optimal contract indeed takes one of the above forms.

Theorem 1. *Suppose that the execution and training routes are equally efficient ($\lambda_S = 2\lambda_D$). There exist thresholds π_F , π_A and π_B such that $\pi_A > \pi_B > \pi_F \equiv 1 + \phi/c$ and the optimal contract can be implemented as follows:*

- (a) *when $\pi > \pi_A$, a training-only contract is optimal;*
- (b) *when $\pi_A > \pi > \pi_B$, there exists a one-switch contract that is optimal;*
- (c) *when $\pi_B > \pi > \pi_F$, an execution-only contract is optimal; and*
- (d) *when $\pi < \pi_F$, the project is infeasible.*

As discussed in Lemma 2, the execution route is preferred when π is lower and the training route is preferred when π is higher. The above theorem aligns with that intuition. In the subsequent subsections, we provide the details of the derivation of this result.

4.2 Promised Utility and Incentive Compatibility

Following the standard approach of the dynamic contract literature, we consider the agent's promised utility as a state variable and write a contract recursively (e.g., [Spear and Srivastava, 1987](#)). For a contract Γ , let $P_0(\Gamma)$ and $U_0(\Gamma)$ be the expected payoffs of the principal and the agent at time 0 when the agent works.

The core of the analysis is the derivation of the principal's value function, denoted by $V(u)$, which represents her maximized expected payoff $P_0(\Gamma)$ subject to the promise-keeping constraint $U_0(\Gamma) = u$ and the incentive compatibility condition, which will be delineated later in this subsection. If a contract Γ satisfies $P_0(\Gamma) = V(u)$ and $U_0(\Gamma) = u$, Γ is said to *implement* a pair of expected payoffs $(V(u), u)$. Once the value function is characterized, the principal solves

$$\bar{u} \equiv \arg \max_{u \geq 0} V(u). \quad (\text{MP})$$

Then, the optimal contract is the contract that implements $(V(\bar{u}), \bar{u})$. In the rest of this subsection, we describe how to derive the value function $V(u)$.

Promised utility upon skill acquisition We begin by considering the principal's problem, given that the advanced skill is acquired at time t . Let u_M^t denote the agent's promised utility, which will be considered as a state variable. Since this case requires only one more breakthrough, it is identical to the single-stage benchmark in [Green and Taylor \(2016a\)](#). They show that the optimal contract is to impose a deadline $t + u_M^t/\phi$ and a linearly diminishing reward schedule $\{\check{R}_s^t\}_{t \leq s \leq t + u_M^t/\phi}$ where

$$\check{R}_s^t = u_M^t + \frac{\phi}{\lambda_S} - \phi(s - t). \quad (4.1)$$

The intuition is that when the agent's promised utility is u_M^t , the principal can incentivize the agent to work at most u_M^t/ϕ units of time. If the principal grants more time without increasing the agent's expected utility, then he will prefer to shirk out the clock.

Incentive compatibility conditions Now consider the agent's problem when the advanced skill has not been acquired. Suppose that the promised utility is u_t at some time t . Under the execution route, if the agent works for a small interval of time $[t, t + dt)$, the breakthrough occurs and the agent receives the reward R_t with a probability $\lambda_D dt$. In this event, however, he loses the continuation utility, thus, the expected payoff from working is $\lambda_D(R_t - u_t)dt$. On the other hand, if he shirks, his payoff is ϕdt . From this, we can derive the incentive compatibility constraint under the execution route ($a_t = 1$):

$$R_t \geq u_t + \frac{\phi}{\lambda_D}. \quad (\text{IC}_1)$$

Next, under the training route, the agent is compensated in the form of a jump in promised utility upon acquiring the advanced skill. Thus, the expected payoff of working for $[t, t + dt)$ is $\lambda_S(u_M^t - u_t)dt$. Then, the incentive compatibility constraint under the training route ($a_t = 0$) is

$$u_M^t \geq u_t + \frac{\phi}{\lambda_S}. \quad (\text{IC}_0)$$

4.3 Value Function Characterization

In this subsection, we characterize the value function of the principal. A natural conjecture is that the principal's expected payoff is maximized when the incentive compatibility conditions bind.¹¹ We outline some key properties of contracts with binding IC conditions, and then characterize the value function.

Deadline and extension With binding IC conditions, the agent's promised utilities should fall at the same rate as the benefit from shirking: $du/dt = \dot{u}_t = -\phi$, or equivalently, $u_t = u_0 - \phi t$. If the problem has not been solved by u_0/ϕ , the promised utility is equal to the agent's outside option 0, thus, the contract is terminated, or equivalently, the

¹¹See Remark 2 for a discussion of the binding IC condition.

deadline of the contract is u_0/ϕ .

When the training route is chosen, to make (IC₀) bind, we have

$$t + \frac{u_M^t}{\phi} = t + \frac{u_t}{\phi} + \frac{1}{\lambda_S} = \frac{u_0}{\phi} + \frac{1}{\lambda_S}.$$

This implies that upon skill acquisition the updated deadline $t + u_M^t/\phi$ extends the original deadline u_0/ϕ by $1/\lambda_S$.

Linearly diminishing rewards Let T denote the deadline u_0/ϕ . By using $u_t = u_0 - \phi t = \phi(T - t)$, to make (IC₁) bind, the reward for solving the problem at time t via the execution route is

$$R_t = u_t + \frac{\phi}{\lambda_D} = \phi \left(T - t + \frac{1}{\lambda_D} \right),$$

which corresponds to R_t^B in Definition 2.

Next, when the skill is acquired at $\check{t}$, to make (IC₀) bind, we have $u_A^{\check{t}} = u_{\check{t}} + \phi/\lambda_S$. Then, by (4.1), the reward for solving the problem at time $t \in [\check{t}, T + 1/\lambda_S]$ via the training route is

$$\check{R}_t^{\check{t}} = u_{\check{t}} + \frac{\phi}{\lambda_S} + \frac{\phi}{\lambda_S} - \phi(t - \check{t}) = \phi \left(T - t + \frac{2}{\lambda_S} \right),$$

which corresponds to R_t^A in Definition 2.

Value function Based on the above observations, we surmise that the principal's value function under agency is linked to the benchmark planner's problem with a deadline and its extension, as explored in the previous section. Let $\mathcal{W}^*(T, \Delta)$ denote the optimal expected social surplus under the deadline T and extension Δ , derived from the optimal policy in Theorem 1. Then, when the agent's promised utility is u , a conjecture for the principal's value function is the expected social surplus from the deadline u/ϕ and the extension $1/\lambda_S$, $\mathcal{W}^*(u/\phi, 1/\lambda_S)$, net of u .

The following proposition verifies this conjecture, with the proof provided in Appendix

B.

Proposition 2. *The principal’s value function $\mathcal{V}$ is characterized as follows:*

$$\mathcal{V}(u) = \mathcal{W}^*(u/\phi, 1/\lambda_S) - u. \quad (4.2)$$

Moreover, $\mathcal{V}$ is concave.

A key step in proving this proposition is finding a contract implementing the pair of the agent’s promised utility, u , and the principal’s expected payoff, $V(u)$. When choosing a path, the principal’s incentives are perfectly aligned with those of the planner—who faces the same deadlines as the principal—in that both want to maximize the expected surplus. Since the planner’s policy with at most one switch is optimal, we show that the principal can implement the pair using a contract that involves at most one switch with linearly diminishing rewards (Appendix B.3).

4.4 Proof of Theorem 1

Now that we have characterized the principal’s value function, the next step is to pin down the optimal initial promised utility level for the agent, $\bar{u}$, which is the solution to (MP). This will establish the starting point of the contract in Figure 4 and determine the deadline length, $\bar{u}/\phi$. The key tradeoff derives from the fact that the rents needed to keep the agent from shirking grow linearly with the deadline, whereas the marginal benefit from extending the deadline falls because the problem is increasingly likely to be solved before the deadline is reached.

Recall that the basic skill is never employed when the extension is greater than $\bar{\Delta}$ (Lemma 1 (a)) and the cutoff is decreasing in π (Lemma 2). Let π_A be the solution of $\bar{\Delta} = 1/\lambda_S$. Then, for all $\pi > \pi_A$, the execution route will not be employed, even near the deadline. This establishes Theorem 1 (a) and is illustrated in Figure 4 (c).

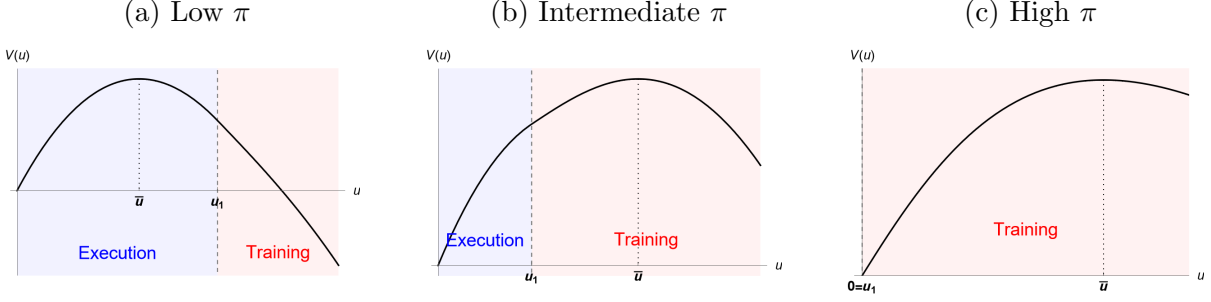


Figure 4: Value functions when routes are equally efficient

When $\pi < \pi_A$, Theorem 1 (b) indicates that the optimal approach is switched from the training route to the execution route when $\hat{T}$ units of time remain. In terms of the promised utility, the switch happens at $\hat{u}_1 \equiv \phi \hat{T}$. Then, the form of the optimal contract depends on whether $\bar{u}$ is greater than $\hat{u}_1$ or not. For example, the value functions in Figure 4a and 4b both involve a switching point $\hat{u}_1$, however, $\bar{u}$ is greater than $\hat{u}_1$ in Figure 4a and less than $\hat{u}_1$ in Figure 4b. Thus, the optimal contracts are an execution-only contract in Figure 4a and a contract with a switch from the training route to the execution route in Figure 4b. Let π_B be the threshold for the project return where the optimal promised utility $\bar{u}$ is equal to the switching point $\hat{u}_1$. Then, for any $\pi \in (\pi_B, \pi_A)$, the optimal contract will involve a switch, establishing Theorem 1 (b). Conversely, when $\pi < \pi_B$, only the execution route will appear in the optimal contract.

Last, the feasibility of the project depends on whether $\bar{u}$ is greater than 0 or not. When $\bar{u}$ is equal to zero, the principal's expected payoff is maximized at $u = 0$, meaning it is optimal for the principal not to initiate the contract—the project is infeasible. This occurs when the principal's flow profit is negative near the deadline T . Since the promised utility u is close to zero near the deadline, the reward R is approximately ϕ/λ_D . Then, the principal's flow profit in $[T - dt, T]$ is approximately

$$\lambda_D dt \cdot \left(\Pi - \frac{\phi}{\lambda_D} \right) - c dt = \lambda_D \left(\Pi - \frac{\phi + c}{\lambda_D} \right) dt.$$

Therefore, the project is feasible if and only if $\pi = \lambda_D \Pi / c$ is greater than $\pi_F \equiv 1 + \phi/c$. This

makes sense because the principal must pay both the operating cost c and (because incentive compatibility binds) the shirking benefit ϕ for the potential duration of the contract. We show that $\pi_B \in (\pi_F, \pi_A)$ (Lemma 10). Then, when $\pi \in (\pi_F, \pi_B)$, the execution-only contract is optimal (Theorem 1 (c)); and when $\pi < \pi_F$, the project becomes infeasible (Theorem 1 (d)).

Remark 1. A mixture of contracts also generates another contract. For example, a contract with a soft deadline—randomly terminating the agent after reaching the soft deadline, as in Green and Taylor (2016a)—can be represented by a mixture of two contracts defined here. However, a mixed contract cannot improve upon the one characterized above. This follows because the value function $\mathcal{V}$ is concave (Lemma 8 (c)).

Consider a set of contracts $\{\Gamma_i\}_{1 \leq i \leq n}$ where the agent’s expected utility under Γ_i is u_i , and the weight is w_i with $\sum_{i=1}^n w_i = 1$ and $\sum_{i=1}^n w_i \cdot u_i = u$. The principal’s expected payoff from this mixture is $\sum_{i=1}^n w_i \cdot P_0(\Gamma_i)$ and the agent’s expected utility is u . By concavity, we have $V(u) \geq \sum_{i=1}^n w_i V(u_i)$. Additionally, $V(u_i) \geq P_0(\Gamma_i)$ holds for all $1 \leq i \leq n$ because $V(u_i)$ is the principal’s maximized expected profit given that the agent’s expected payoff is u_i . Thus, $V(u)$ is greater than or equal to the expected payoff of the mixed contract. Hence, any mixed contract cannot improve upon the deterministic contract characterized above.

Remark 2. In Green and Taylor (2016b), there is a parametric region where the incentive compatibility constraint does not bind. This occurs because the deadline extension set by binding IC is not long enough to make the probability of two breakthroughs sufficiently high. In contrast, in this model, when such a situation arises, the principal can switch to the execution path, which is appealing enough to replace the training path. This ensures that IC is always binding.

5 Extension: Efficiency Loss from Training

5.1 Efficiency Loss from Staging

Proposition 1 neutralizes efficiency, so that the choice between the routes is driven by the deadline alone. We now allow staging to be the less efficient route and ask what survives. Write

$$\eta \equiv \frac{\lambda_S}{\lambda_D} - 1, \quad (5.1)$$

so that the rate restriction maintained in Section 2 is exactly $0 < \eta \leq 1$: staging is technologically innocuous when $\eta = 1$, and $\eta < 1$ measures the efficiency loss it imposes.

The answer is that one force is added and one phase with it. Because the staged route is now slower in total, it is dominated when the deadline is remote, so the direct route reappears at the far end of the horizon. The optimal policy switches at most twice, and the order of its phases is pinned down.

Definition 3. A policy is called a *two-switch policy* if there exist intermediate deadlines $0 \leq S_1 \leq S_2 \leq T$ such that (i) the principal chooses the direct route up to S_1 ($a_t = 1$ for $t \leq S_1$), (ii) she chooses the staged route on $(S_1, S_2]$ ($a_t = 0$), (iii) if the interim milestone is delivered before S_2 , she works toward completion until the extended deadline $T + \Delta$, and (iv) if the milestone is not delivered by S_2 , she switches back to the direct route until the deadline T .

This class nests every policy in Proposition 1. A two-switch policy with $S_1 = 0$ is a one-switch policy with intermediate deadline S_2 , and hence the staged-only policy when in addition $S_2 = T$; a two-switch policy with $S_1 = S_2$ is the direct-only policy. What is new is the case $0 < S_1 < S_2 < T$, in which the direct route is used both at the beginning and at the end of the horizon and staging occupies an interior window.

The extension cutoff of Proposition 1 generalizes to

$$\bar{\Delta}(\eta, \pi) \equiv \begin{cases} \frac{1}{\lambda_S} \log \left[\frac{(1+\eta)\pi - 1}{\eta\pi - 1} \right], & \text{if } \eta\pi > 1, \\ \infty, & \text{otherwise,} \end{cases} \quad (5.2)$$

which is the cutoff in Proposition 1 at $\eta = 1$. Writing the cutoff this way makes explicit how it depends on the two primitives that matter, the efficiency of staging and the project return.

Proposition 3 (Optimal policy under efficiency loss). *Suppose that staging is less efficient ($0 < \eta < 1$). There exist $\hat{T}$ and $\check{T}$ with $0 \leq \hat{T} \leq \check{T} \leq \infty$, both independent of T , such that for every deadline T and every extension $\Delta \geq 0$ the two-switch policy with*

$$S_1 = \max\{T - \check{T}, 0\} \quad \text{and} \quad S_2 = \max\{T - \hat{T}, 0\}$$

is optimal, with the convention that $S_i = 0$ when the corresponding threshold is infinite. Moreover:

- (a) **(Long extension)** *if $\Delta > \bar{\Delta}(\eta, \pi)$, then $\hat{T} = 0 < \check{T} < \infty$: the staged-only policy is optimal when $T \leq \check{T}$, and when $T > \check{T}$ the direct route is used early rather than late;*
- (b) **(Short extension)** *if $\Delta \leq \bar{\Delta}(\eta, \pi)$, set*

$$x^* \equiv \left[\frac{\eta(\pi - 1)}{(1 + \eta)[(1 + \eta)\pi - 1]} \cdot e^{\lambda_S \Delta} \right]^{\frac{1}{\eta}}. \quad (5.3)$$

If $x^ \geq 1$ or $L(\eta, \pi, \Delta) \leq 0$, where*

$$L(\eta, \pi, \Delta) \equiv [(1 + \eta)\pi - 1] (1 - e^{-\lambda_S \Delta} (x^*)^{1+\eta}) - \pi - \eta(\pi - 1)(1 - x^*), \quad (5.4)$$

then $\hat{T} = \check{T} = \infty$ and the direct-only policy is optimal for every T . Otherwise $0 \leq \hat{T} <$

$\check{T} < \infty$, so that the direct-only policy is optimal when $T \leq \hat{T}$, the one-switch policy with intermediate deadline $T - \hat{T}$ is optimal when $\hat{T} < T \leq \check{T}$, and a two-switch policy with both switches interior is optimal when $T > \check{T}$.

Unlike Proposition 1(a), part (a) here requires the strict inequality $\Delta > \bar{\Delta}$. At $\Delta = \bar{\Delta}$ the two routes are exactly indifferent at the deadline, and which one is used just inside it turns on the sign of $\eta^2\pi - 1$; this quantity is strictly positive whenever $\eta = 1$, which is why the boundary case can be absorbed into part (a) under technologically innocuous staging but not here.¹²

The proposition says that the deadline effect and the efficiency effect operate on different parts of the horizon, and that they can be read off the remaining time rather than the calendar. When little time is left, the deadline effect dominates and the direct route is used, exactly as in Proposition 1: needing one slow success beats needing two fast ones. As the remaining time grows past $\hat{T}$, the deadline effect reverses and staging takes over. But the deadline effect washes out as the horizon lengthens, while the efficiency effect does not: with an unlimited horizon the two routes deliver $\Pi - c/\lambda_D$ and $\Pi - 2c/\lambda_S$, and the second is smaller precisely when $\eta < 1$. So beyond $\check{T}$ efficiency wins and the direct route returns. Under Proposition 1 this last phase is absent because the two limits coincide; formally, $\check{T} \rightarrow \infty$ as $\eta \uparrow 1$, and Proposition 3 converges to Proposition 1.

Part (a) records a further asymmetry. When the extension is generous, staging is attractive near the deadline, since delivering the milestone buys Δ additional units of time exactly when time is scarce. The direct route is then used early rather than late, reversing the ordering of the one-switch policy in Proposition 1(b) — an ordering that cannot arise when staging is technologically innocuous.

¹²The relevant object is $\Phi(0)$, defined in (B.19), evaluated at $\Delta = \bar{\Delta}$; a short computation gives $\Phi(0) = \lambda_D c(\eta^2\pi - 1)$. When this is negative the boundary case falls into part (b) with $x^* \geq 1$, so the direct-only policy is optimal for every T .

We last consider the case where training generates an efficiency loss, i.e., $\lambda_S < 2\lambda_D$. This introduces efficiency as another economic force, alongside milestone and deadline effects, that shape the optimal contract.

We define a parameter $\eta \equiv \lambda_S/\lambda_D - 1$, which measures the relative efficiency of the training path. Note that $0 < \eta < 1$, and the efficiency loss increases as η decreases. In this section, we characterize the optimal contracts for two cases: (i) when the efficiency loss is small ($\eta > \bar{\eta} \equiv \max\{\sqrt{c/(c+\phi)}, 1/(e-1)\}$); and (ii) when the efficiency loss is large ($\eta < \underline{\eta} \equiv \min\{c/(c+\phi), 1/(e-1)\}$).¹³

In Figures 5 and 6, we illustrate the principal's value functions when there are efficiency losses from training. A key characteristic of these value functions is that the execution route is employed when the promised utility is high, indicating that the deadline is far off. To understand these dynamics, we compare the execution-only and training-only contracts again. As time horizons become longer, the sums of expected payoffs for both players from these contracts converge to the expected surpluses of the no-deadline benchmark: $\Pi - c/\lambda_D$ for the execution-only contract and $\Pi - 2c/\lambda_S$ for the training-only contract. Therefore, efficiency determines which approach should be chosen. Since we focus on the case where the training route is less efficient than the execution route, the principal would choose the execution route when the deadline is distant.

This observation, combined with milestone and deadline effects discussed in the previous sections, leads us to conjecture that there will be two switching points $\hat{u}_1$ and $\hat{u}_2$ in determining the value function. The execution route is chosen when $u > \hat{u}_2$ or $u < \hat{u}_1$, and the training route is chosen when $u \in (\hat{u}_1, \hat{u}_2)$.

Small efficiency loss When the efficiency loss is relatively small, we show that $\hat{u}_2$ is always greater than the optimal initial promised utility level $\bar{u}$ (Lemma OA.9). It implies that the switch occurs at most once in the optimal contract. Therefore, a result similar to

¹³These do not cover cases where the efficiency loss is intermediate. In such cases, the form of the optimal contract depends heavily on the parameter values η and Π , resulting in many subcases to analyze. Thus, we focus on the extreme cases to provide results with clear economic implications.

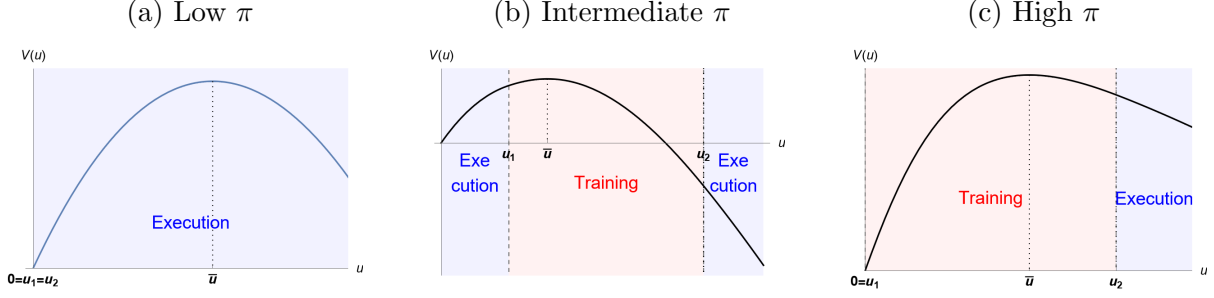


Figure 5: Value functions when the efficiency loss is small

the no-efficiency-loss case holds. In other words, Theorem 1 is robust to small efficiency losses.

Proposition 4. *Suppose that $\eta \in (\bar{\eta}, 1)$, i.e., the efficiency loss from training is small. There exist thresholds $\tilde{\pi}_A(\eta)$ and $\tilde{\pi}_B(\eta)$ with $\tilde{\pi}_A(\eta) > \tilde{\pi}_B(\eta) > \pi_F$ such that the optimal contract is determined as follows:*

- (a) *when $\pi > \tilde{\pi}_A(\eta)$, a training-only contract is optimal;*
- (b) *when $\tilde{\pi}_A(\eta) > \pi > \tilde{\pi}_B(\eta)$, there exists a one-switch contract that is optimal;*
- (c) *when $\tilde{\pi}_B(\eta) > \pi > \pi_F$, an execution-only contract is optimal.*

Large efficiency loss Now suppose that the efficiency loss is large. Figure 6 illustrates that the training path is either not employed at all (for small π) or is employed in the middle of the contract (for large π). As π increases, the milestone effect becomes more significant, as it actuates the monitoring ability of the principal. However, the execution approach is preferred at the beginning of the contract due to its efficiency and at the end of the contract due to the deadline effect. Therefore, if training is ever employed, the contract will involve two switches. We formally define the two-switch contract and state the theorem for the case of large efficiency loss.

Definition 4. A contract is called a *two-switch contract* with a final deadline T and two intermediate deadlines $0 < S_1 < S_2 < T$ if

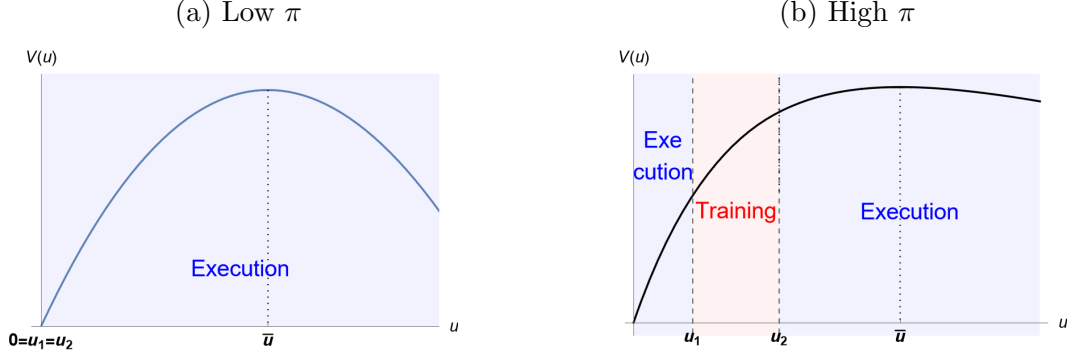


Figure 6: Value functions when the efficiency loss is large

- (i) the agent is recommended to solve the problem using the basic skill by S_1 and the reward upon project completion at time t is R_t^B ,
- (ii) if the problem is not solved by S_1 , the agent is recommended to acquire the advanced skill by S_2 , and if the advanced skill is acquired before S_2 , the contract is extended by $1/\lambda_S$ with the reward upon solution at time t being R_t^A ,
- (iii) if the advanced skill is not acquired by S_2 , the agent is recommended to solve the problem using the basic skill by T and the reward upon project completion at time t is R_t^B , and
- (iv) the contract is terminated if the project is not completed by T .¹⁴

Proposition 5. *Suppose that η is less than $\underline{\eta}$, i.e., the efficiency loss from training is large. There exists a threshold $\tilde{\pi}_M(\eta)$ with $\tilde{\pi}_M(\eta) > \pi_F$ such that the optimal contract is determined as follows:*

- (a) *when $\pi > \tilde{\pi}_M(\eta)$, there exists a two-switch contract that is optimal;*
- (b) *when $\tilde{\pi}_M(\eta) > \pi > \pi_F$, an execution-only contract is optimal.*

Notably, the principal typically prefers the execution route, since training entails a substantial efficiency loss. When π is sufficiently large, however, she may exploit the monitoring

¹⁴The rewards R_t^A and R_t^B are defined in the same way as in the one-switch contract.

benefit from the milestone effect by choosing the training path. If monitoring occurs, it is optimal to place it in the middle of the contract: efficiency dominates at the start, while the deadline effect dominates near the end.

For high-return projects, the theorem shows that all three forces—efficiency, the milestone effect, and the deadline effect—shape the contract. The principal begins with recommending execution with the basic skill (i.e., efficiency is the primary concern). If the problem remains unsolved by a certain time, she recommends acquiring the advanced skill to monitor the agent and bring them closer to the solution (i.e., the milestone effect becomes the primary concern). She extends the deadline if the agent acquires the advanced skill, but if he does not acquire it until the deadline is near, the principal reverts to execution with the basic skill in a “last-ditch” attempt to solve the problem (i.e., the deadline effect becomes the preeminent motivation).

6 Conclusion

In this article, we study the economic tradeoffs between execution and training in solving a problem in the presence of agency frictions. The optimal contract is determined by the interplay of three effects: efficiency, milestone, and endogenous deadlines. We show that the form of the optimal contract crucially depends on the project return. When the efficiency loss from training does not exist or is small, the optimal contract involves at most one switch. Specifically, if the project return is low, the principal always recommends the agent to solve the problem with a basic skill, whereas if the project return is high, the principal always recommends the agent to acquire the advanced skill, then extend the deadline upon skill acquisition. If the project return is intermediate, it is optimal to begin with training and then switch to execution upon lack of skill acquisition. When the efficiency loss is large, the principal generally recommends the agent to solve a problem with the more efficient basic skill route. However, if the project return is above a certain cutoff, for a short period of time

in the middle of the contract, the principal recommends the agent to acquire the advanced skill to mitigate moral hazard (i.e., there may be two switches).

There are numerous avenues open for further research. For example, the principal may be able to design the approaches directly. In this article, we assume that the two approaches are exogenously given and the principal chooses between them. However, in practice, a project manager often designs how many milestones to partition the main project into and how difficult each subproject is. We could also introduce ‘learning by doing’ into the model. If we assume that the agent learns from early errors, the arrival rate of project completion would increase over time.¹⁵ We leave these intriguing questions—and others—for future work.

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¹⁵This possibility contrasts with the setting considered by [Carnehl and Schneider \(2023\)](#), where learning causes the expected arrival rate to fall.

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Appendix

A Proofs for Section 3

A.1 Expected Surpluses of Benchmark Policies

In this section, we provide formal representations of the expected social surpluses under the direct-only and staged-only policies.

Suppose that the principal employs the direct-only policy. Recall that the completion probability $\mathcal{P}^D(T)$ is derived in (3.1). Next, the expected duration of project completion is

$$\mathcal{D}^D(T) \equiv \int_0^T \tau \cdot \lambda_D \cdot e^{-\lambda_D \tau} d\tau + T \cdot e^{-\lambda_D T} = \frac{1}{\lambda_D} (1 - e^{-\lambda_D T}). \quad (\text{A.1})$$

Then, the expected social surplus of the direct-only policy is

$$\mathcal{W}^D(T) \equiv \Pi \cdot \mathcal{P}^D(T) - c \cdot \mathcal{D}^D(T) = \left(\Pi - \frac{c}{\lambda_D} \right) \cdot (1 - e^{-\lambda_D T}). \quad (\text{A.2})$$

Now suppose that the principal employs the staged-only policy. Recall that the completion probability $\mathcal{P}^S(T, \Delta)$ is derived in (3.2). Conditional on milestone delivery at τ_m , the expected duration is

$$\mathcal{D}^1(T, \Delta, \tau_m) \equiv \int_{\tau_m}^{T+\Delta} \tau \cdot \lambda_S e^{-\lambda_S(\tau-\tau_m)} d\tau + (T + \Delta) \cdot e^{-\lambda_S(T+\Delta-\tau_m)}.$$

Then, the total expected duration of the project can be derived as follows:

$$\begin{aligned} \mathcal{D}^S(T, \Delta) &\equiv \int_0^T \mathcal{D}^1(T, \Delta, \tau_m) \cdot \lambda_S e^{-\lambda_S \tau_m} d\tau_m + T \cdot e^{-\lambda_S T} \\ &= \frac{2}{\lambda_S} (1 - e^{-\lambda_S T}) - T \cdot e^{-\lambda_S(T+\Delta)}. \end{aligned} \quad (\text{A.3})$$

The expected social surplus of the staged-only policy is thus

$$\begin{aligned}\mathcal{W}^S(T, \Delta) &\equiv \Pi \cdot \mathcal{P}^S(T, \Delta) - c \cdot \mathcal{D}^S(T, \Delta) \\ &= \left(\Pi - \frac{2c}{\lambda_S} \right) \cdot (1 - e^{-\lambda_S T}) - \lambda_S \left(\Pi - \frac{c}{\lambda_S} \right) \cdot T \cdot e^{-\lambda_S(T+\Delta)}.\end{aligned}\tag{A.4}$$

A.2 Proof of Lemma 1

Proof of Lemma 1. Observe that $\mathcal{P}^D(T, 0) \geq \mathcal{P}^S(T, 0)$ is equivalent to:

$$(1 + \lambda_S T) \geq e^{(\lambda_S - \lambda_D)T}.$$

Equality holds at $T = 0$. The left-hand side linearly increases with slope λ_S , while the right-hand side exponentially increases with an initial slope at $T = 0$ of $\lambda_S - \lambda_D \in (0, \lambda_S)$. Therefore, for sufficiently small T , $\mathcal{P}^D(T, 0) > \mathcal{P}^S(T, 0)$, but there exists a unique cutoff $\check{T} > 0$ equating the two sides. Thus, $\mathcal{P}^D(T, 0) > \mathcal{P}^S(T, 0)$ for all $T < \check{T}$, and $\mathcal{P}^D(T, 0) < \mathcal{P}^S(T, 0)$ for all $T > \check{T}$. $\square$

A.3 Proof of Proposition 1

Value functions and the HJB equation Let W_x^1 denote the expected surplus once the interim milestone has been delivered and x units of time remain. By the same steps that deliver (A.2),

$$W_x^1 \equiv \left(\Pi - \frac{c}{\lambda_S} \right) \cdot (1 - e^{-\lambda_S x}).\tag{A.5}$$

If the milestone arrives with z units of time remaining, the deadline extends by Δ , so the continuation surplus is $W_{z+\Delta}^1$.

Before the milestone arrives, the optimal expected surplus $W_z^{0,\Delta}$ with z units of time

remaining satisfies

$$\dot{W}_z^{0,\Delta} = \max_{a_z \in \{0,1\}} [\lambda_D a_z \cdot (\Pi - W_z^{0,\Delta}) + \lambda_S (1 - a_z) \cdot (W_{z+\Delta}^1 - W_z^{0,\Delta}) - c], \quad (\text{HJB}_W)$$

with $W_0^{0,\Delta} = 0$, since the project is abandoned at the deadline. By standard verification arguments (e.g., Proposition 3.2.1 in Bertsekas, 1995), if $w : [0, T] \rightarrow \mathbb{R}$ is differentiable, satisfies (HJB_W) , and has $w(0) = 0$, then $w(z) = W_z^{0,\Delta}$.

The staged advantage For a continuation value W , define the net flow advantage of the staged route over the direct route,

$$\Lambda(z; W) \equiv \lambda_S (W_{z+\Delta}^1 - W) - \lambda_D (\Pi - W) = \lambda_S W_{z+\Delta}^1 - \lambda_D \Pi - (\lambda_S - \lambda_D)W. \quad (\text{A.6})$$

The staged route attains the maximum in (HJB_W) at z if and only if $\Lambda(z; W_z^{0,\Delta}) \geq 0$. Two observations organize the proof. First, because $\lambda_S > \lambda_D$, the map $W \mapsto \Lambda(z; W)$ is strictly decreasing: a higher continuation value makes the direct route relatively more attractive, since the direct route delivers Π outright whereas the staged route only advances the project by one stage. Second, and consequently, Λ must be evaluated at the value function that actually prevails at z . Substituting the direct-only value $\mathcal{W}^D(z)$ is legitimate only where the direct route is in fact used, and it *overstates* the staged advantage elsewhere. Lemma 3 locates the switch point using $\mathcal{W}^D$, and Lemma 4 then propagates the comparison beyond it using the correct value function.

The one-switch value function Fix a switching threshold $Z \in [0, T]$ in remaining time: the direct route is used when $z < Z$ and the staged route when $z \geq Z$, which is the one-switch policy with intermediate deadline $S = T - Z$. Let $\mathcal{W}_Z^{SD}(z, \Delta)$ denote its expected

surplus with z units of time remaining. It satisfies

$$\dot{\mathcal{W}}_Z^{SD}(z, \Delta) = \lambda_D \cdot (\Pi - \mathcal{W}_Z^{SD}(z, \Delta)) - c \quad \text{for } z < Z, \quad (\text{A.7})$$

$$\dot{\mathcal{W}}_Z^{SD}(z, \Delta) = \lambda_S \cdot (W_{z+\Delta}^1 - \mathcal{W}_Z^{SD}(z, \Delta)) - c \quad \text{for } z \geq Z, \quad (\text{A.8})$$

and solving with $\mathcal{W}_Z^{SD}(0, \Delta) = 0$ gives

$$\mathcal{W}_Z^{SD}(z, \Delta) = \begin{cases} \mathcal{W}^D(z), & \text{if } z \leq Z, \\ \left(\Pi - \frac{2c}{\lambda_S} \right) \cdot (1 - e^{-\lambda_S(z-Z)}) + \mathcal{W}^D(Z) \cdot e^{-\lambda_S(z-Z)} & \text{if } z > Z. \\ -\lambda_S \left(\Pi - \frac{c}{\lambda_S} \right) \cdot (z - Z) \cdot e^{-\lambda_S(z+\Delta)}, & \end{cases} \quad (\text{A.9})$$

The two benchmark policies are the endpoints of this family: $\mathcal{W}_0^{SD}(z, \Delta) = \mathcal{W}^S(z, \Delta)$ is the staged-only surplus and $\mathcal{W}_T^{SD}(z, \Delta) = \mathcal{W}^D(z)$ is the direct-only surplus. We prove the proposition by exhibiting a $Z \in [0, T]$ for which $\mathcal{W}_Z^{SD}(\cdot, \Delta)$ solves (HJB_W).

Optimal route at the deadline Since $W_0^{0,\Delta} = 0$, the staged route is preferred at the deadline if and only if $\Lambda(0; 0) \geq 0$, that is,

$$\lambda_D \Pi \leq (\lambda_S \Pi - c) \cdot (1 - e^{-\lambda_S \Delta}). \quad (\text{A.10})$$

Under $\lambda_S = 2\lambda_D$, dividing by c and rearranging shows that (A.10) holds if and only if $\Delta \geq \bar{\Delta}$.

Lemma 3 (Locating the switch point). *Suppose $\lambda_S = 2\lambda_D$ and $\Delta < \bar{\Delta}$. Then there exists $\hat{T} > 0$ such that (i) $\Lambda(z; \mathcal{W}^D(z)) < 0$ for all $z < \hat{T}$; and (ii) $\Lambda(\hat{T}; \mathcal{W}^D(\hat{T})) = 0$.*

Proof of Lemma 3. Substituting (A.5) and $\mathcal{W}^D(z) = (\Pi - c/\lambda_D)(1 - e^{-\lambda_D z})$ into (A.6) and writing $x \equiv e^{-\lambda_D z} \in (0, 1]$,

$$\Lambda(z; \mathcal{W}^D(z)) = (\lambda_S \Pi - c) (1 - e^{-\lambda_S \Delta} x^{\lambda_S/\lambda_D}) - \lambda_D \Pi - (\lambda_S - \lambda_D) \left(\Pi - \frac{c}{\lambda_D} \right) (1 - x). \quad (\text{A.11})$$

Under $\lambda_S = 2\lambda_D$ the exponent λ_S/λ_D equals 2 and the constant terms cancel, so (A.11) reduces to the quadratic

$$\tilde{\Lambda}(x) \equiv (\lambda_D \Pi - c) \cdot x - (\lambda_S \Pi - c) \cdot e^{-\lambda_S \Delta} \cdot x^2 = x [(\lambda_D \Pi - c) - (\lambda_S \Pi - c)e^{-\lambda_S \Delta}]. \quad (\text{A.12})$$

Since $\pi > 1$ by (2.1), both $\lambda_D \Pi - c > 0$ and $\lambda_S \Pi - c > 0$, so $\tilde{\Lambda}$ is strictly concave with roots 0 and

$$\hat{x} \equiv \frac{\lambda_D \Pi - c}{\lambda_S \Pi - c} \cdot e^{\lambda_S \Delta}. \quad (\text{A.13})$$

Comparing (A.13) with (A.10) shows that $\hat{x} < 1$ precisely when $\Delta < \bar{\Delta}$. Hence $\tilde{\Lambda}(\hat{x}) = 0$ and $\tilde{\Lambda}(x) < 0$ for all $\hat{x} < x \leq 1$.

Set $\hat{T} \equiv -\log(\hat{x})/\lambda_D > 0$. Because $x = e^{-\lambda_D z}$ is strictly decreasing in z , $z < \hat{T}$ is equivalent to $x > \hat{x}$, which yields (i); and $\tilde{\Lambda}(\hat{x}) = 0$ yields (ii). $\square$

Condition (ii) is precisely the indifference condition delivered by smooth pasting: at $\hat{T}$ the two branches of (HJB_W) have equal flow value, so the candidate value function is continuously differentiable there.

Lemma 4 (Propagating past the switch point). *Suppose $\lambda_S = 2\lambda_D$ and $\Lambda(Z; \mathcal{W}^D(Z)) \geq 0$ for some $Z \geq 0$. Then $\Lambda(z; \mathcal{W}_Z^{SD}(z, \Delta)) \geq 0$ for all $z > Z$.*

Proof of Lemma 4. For $z > Z$ write $y \equiv e^{-\lambda_S(z-Z)} \in (0, 1)$ and $\kappa \equiv (\Pi - c/\lambda_S) e^{-\lambda_S(Z+\Delta)} > 0$. Rearranging (A.9) and (A.5) gives

$$\begin{aligned} \mathcal{W}_Z^{SD}(z, \Delta) &= \mathcal{W}^D(Z) + \left(\Pi - \frac{2c}{\lambda_S} - \mathcal{W}^D(Z) \right) (1 - y) + \kappa \log(y) y, \\ W_{z+\Delta}^1 &= W_{Z+\Delta}^1 + \left(\Pi - \frac{c}{\lambda_S} - W_{Z+\Delta}^1 \right) (1 - y). \end{aligned}$$

Substituting these into (A.6) and collecting terms, $\Lambda(z; \mathcal{W}_Z^{SD}(z, \Delta)) = \Lambda_2(y)$, where

$$\Lambda_2(y) \equiv \Lambda(Z; \mathcal{W}^D(Z)) + h \cdot (1 - y) - (\lambda_S - \lambda_D) \kappa \log(y) y, \quad (\text{A.14})$$

$$h \equiv \lambda_S \left(\Pi - \frac{c}{\lambda_S} - W_{Z+\Delta}^1 \right) - (\lambda_S - \lambda_D) \left(\Pi - \frac{2c}{\lambda_S} - \mathcal{W}^D(Z) \right).$$

Note that $\mathcal{W}_Z^{SD}$ enters (A.6) with coefficient $-(\lambda_S - \lambda_D)$, which is what the final term of (A.14) records. Differentiating twice,

$$\Lambda_2''(y) = -\frac{(\lambda_S - \lambda_D) \kappa}{y} < 0,$$

so Λ_2 is strictly concave on $(0, 1)$. At the upper endpoint, $\Lambda_2(1) = \Lambda(Z; \mathcal{W}^D(Z)) \geq 0$ by hypothesis. At the lower endpoint, $\log(y)y \rightarrow 0$ and

$$\lim_{y \rightarrow 0} \Lambda_2(y) = \Lambda(Z; \mathcal{W}^D(Z)) + h = \frac{\lambda_S - 2\lambda_D}{\lambda_S} c = 0, \quad (\text{A.15})$$

where the final equality uses $\lambda_S = 2\lambda_D$. Strict concavity together with $\Lambda_2(1) \geq 0$ and $\lim_{y \rightarrow 0} \Lambda_2(y) = 0$ implies $\Lambda_2(y) \geq 0$ for all $y \in (0, 1)$, which is the claim. $\square$

Equation (A.15) is where technologically innocuous staging does its work: the two endpoint values of Λ_2 are non-negative only because the efficiency wedge $\lambda_S - 2\lambda_D$ vanishes.

Proof of Proposition 1. (a) **Long extension** ($\Delta \geq \bar{\Delta}$). Since $\mathcal{W}^D(0) = 0$, (A.10) gives $\Lambda(0; \mathcal{W}^D(0)) \geq 0$. Applying Lemma 4 with $Z = 0$ yields $\Lambda(z; \mathcal{W}^S(z, \Delta)) \geq 0$ for all $z > 0$, so the staged branch attains the maximum in (HJB_W) at every z . By (A.8), $\mathcal{W}^S(\cdot, \Delta) = \mathcal{W}_0^{SD}(\cdot, \Delta)$ therefore solves (HJB_W) on $\mathbb{R}_+$ with the correct boundary condition, and the staged-only policy is optimal.

(b) **Short extension** ($\Delta < \bar{\Delta}$). Let $\hat{T}$ be as in Lemma 3.

If $T < \hat{T}$, then Lemma 3(i) gives $\Lambda(z; \mathcal{W}^D(z)) < 0$ for all $z \in [0, T]$, so the direct branch attains the maximum throughout. By (A.7), $\mathcal{W}^D$ solves (HJB_W) on $[0, T]$ and the direct-only policy is optimal.

If $T \geq \hat{T}$, consider $\mathcal{W}_T^{SD}$. For $z \leq \hat{T}$ it coincides with $\mathcal{W}^D(z)$, and Lemma 3(i) shows the direct branch attains the maximum there. At $z = \hat{T}$, Lemma 3(ii) delivers both value

matching and smooth pasting, so $\mathcal{W}_{\hat{T}}^{SD}$ is continuously differentiable at $\hat{T}$. That same condition supplies the hypothesis of Lemma 4 at $Z = \hat{T}$, whence $\Lambda\left(z; \mathcal{W}_{\hat{T}}^{SD}(z, \Delta)\right) \geq 0$ and the staged branch attains the maximum for all $z > \hat{T}$. Hence $\mathcal{W}_{\hat{T}}^{SD}$ is differentiable, satisfies (HJB_W) on $[0, T]$, and vanishes at $z = 0$; by the verification argument it is the value function, and the one-switch policy with intermediate deadline $S = T - \hat{T}$ is optimal.

□

B Proofs for Section 4

B.1 Expected Payoffs

In this section, we formally present the expected payoffs of the principal and the agent, conditional on the agent's effort schedule.

Post-skill-acquisition payoffs We begin with the subgame where the advanced skill is acquired at time t . Let u_M^t denote the agent's continuation utility—the expected payoff when the agent works until the problem is solved or the deadline T^t is reached.¹⁶

Since this subgame only requires one more breakthrough, it is identical to the single-stage benchmark of Green and Taylor (2016a). They show that the principal's value function V_M is characterized as follows:

$$V_M(u_M^t) = W_{u_M^t/\phi}^1 - u_M^t = \left(\Pi - \frac{c}{\lambda_S}\right) \left(1 - e^{-\frac{\lambda_S}{\phi} u_M^t}\right) - u_M^t. \quad (\text{B.1})$$

Expected payoffs at time 0 Now consider the problem at time 0. The agent's admissible effort schedule (prior to either solution or skill acquisition) is $\mathcal{B} \equiv \{\{\beta_t\}_{0 \leq t \leq T} : \beta_t \in \{0, 1\}\}$. In this case, any arrival depends not only on the agent's effort (β) but also the principal's

¹⁶Specifically, the updated contract $\hat{\Gamma}^t$ determines

path choice ($a = \{a_t\}_{0 \leq t \leq T}$). Given (a, β) , the probability that neither the solution nor the advanced skill has arrived by time τ is $f(a, \beta; 0, \tau)$, where

$$f(a, \beta; t, \tau) \equiv e^{-\lambda_D \int_t^\tau a_s \beta_s ds - \lambda_S \int_t^\tau (1-a_s) \beta_s ds}.$$

Accordingly, the probability density of the solution at time τ is $\lambda_D a_\tau \beta_\tau \cdot f(a, \beta; 0, \tau)$ and that of the skill acquisition at time τ is $\lambda_S (1 - a_\tau) \beta_\tau \cdot f(a, \beta; 0, \tau)$. If the agent follows the recommendation, the above expressions simplify to: $\tilde{f}(a; t, \tau) \equiv e^{-\lambda_D \int_t^\tau a_s ds - \lambda_S \int_t^\tau (1-a_s) ds}$, $\lambda_D a_\tau \cdot \tilde{f}(a; t, \tau)$, and $\lambda_S (1 - a_\tau) \cdot \tilde{f}(a; t, \tau)$.

Given a contract Γ , an effort schedule β and under the assumption that the agent follows the recommendation in the updated contract, the principal's expected payoff at time 0 is:

$$P_0(\beta, \Gamma) \equiv \int_0^T \{(\Pi - R_\tau) \cdot \lambda_D a_\tau \beta_\tau + V_M(u_M^\tau) \cdot \lambda_S (1 - a_\tau) \beta_\tau - c\} \cdot f(a, \beta; 0, \tau) d\tau.$$

while the agent's expected payoff at time t is:

$$U_0(\beta, \Gamma) \equiv \int_0^T \{R_\tau \cdot \lambda_D a_\tau \beta_\tau + u_M^\tau \cdot \lambda_S (1 - a_\tau) \beta_\tau + \phi(1 - \beta_\tau)\} \cdot f(a, \beta; 0, \tau) d\tau.$$

Similarly, the expected payoffs of the principal and the agent when the agent follows the recommendation can be obtained by evaluating the general expressions using $\tilde{f}(a; 0, \tau)$ in place of $f(a, \hat{\beta}^t; 0, \tau)$. We denote these by $\tilde{P}_0(\Gamma)$ and $\tilde{U}_0(\Gamma)$, respectively.

Incentive Compatibility Using the terms defined above, we define incentive compatibility (IC) of contracts as follows.

Definition 5. A contract $\Gamma = \{T, \{a_t, R_t, \hat{\Gamma}^t\}_{0 \leq t \leq T}\}$ is *incentive compatible* if $\tilde{U}_0(\Gamma) \geq U_0(\beta, \Gamma)$ for all $\beta \in \mathcal{B}$.

The objective of the principal is to find a contract Γ that maximizes her ex ante expected payoff $P_0(\Gamma)$ subject to the incentive compatibility constraint and the limited liability con-

straints $R_\tau \geq 0$ for all $\tau \in [0, T]$ and $R_\tau^t \geq 0$ for all $t \in [0, T]$ and $\tau \in [t, T^t]$. Designate such a contract as an *optimal contract*.

B.2 Recursive Formulation

B.2.1 The Agent's Problem

Given a contract Γ , define the continuation utility that the advanced skill is acquired at t as $u_M^t = \tilde{U}^t(\hat{\Gamma}^t)$ and the continuation utility at t when neither the problem is solved nor the advanced skill is acquired as

$$u_t = \tilde{U}_t(\Gamma) \equiv \int_t^T \{R_\tau \cdot \lambda_D a_\tau + u_M^t \cdot \lambda_S (1 - a_\tau)\} \cdot \tilde{f}(a; t, \tau) d\tau.$$

Observe that

$$0 = \dot{u}_t + (R_t - u_t)\lambda_D a_t + (u_M^t - u_t)\lambda_S (1 - a_t) \quad (\text{HJB}_{PK})$$

where $\dot{u}_t \equiv \frac{du_t}{dt}$.

Also note that (IC₀) for $a_t = 0$ and (IC₁) for $a_t = 1$ can be written together as follows:

$$(R_t - u_t)\lambda_D a_t + (u_M^t - u_t)\lambda_S (1 - a_t) \geq \phi. \quad (\text{IC})$$

The following lemma shows that this condition serves as a sufficient condition for incentive compatibility defined in Definition 5.

Lemma 5. *Given a contract Γ , suppose that there exists a continuous and differentiable process $\{u_t\}_{0 \leq t \leq T}$ satisfying (HJB_{PK}) and $u_T = 0$, and (IC) holds for $0 \leq t \leq T$. Then, Γ is incentive compatible.*

Proof of Lemma 5. The proof is inspired by Proposition 3.2.1 in Bertsekas (1995).

Consider an arbitrary admissible action $\beta \in \mathcal{B}$. Using (HJB_{PK}), (IC₀) and (IC₁), we can

derive that

$$0 \geq \dot{u}_t + (R_t - u_t)\lambda_D a_t \beta_t + (u_M^t - u_t)\lambda_S(1 - a_t)\beta_t + \phi(1 - \beta_t),$$

or equivalently,

$$-\dot{u}_t + (\lambda_D a_t + \lambda_S(1 - a_t))\beta_t \cdot u_t \geq R_t \cdot \lambda_D a_t \beta_t + u_M^t \cdot \lambda_S(1 - a_t)\beta_t + \phi(1 - \beta_t).$$

It is further equivalent to

$$\frac{d}{dt} [-u_t \cdot f(a, \beta; 0, t)] \geq [(R_t \lambda_D a_t + u_M^t \lambda_S(1 - a_t)) \beta_t + \phi(1 - \beta_t)] \cdot f(a, \beta; 0, t).$$

By integrating the above inequality from 0 to T and using $u_T = 0$, we can derive that

$$u_0 \geq \int_0^T [(R_t \lambda_D a_t + u_M^t \lambda_S(1 - a_t)) \beta_t + \phi(1 - \beta_t)] \cdot f(a, \beta; 0, t) dt = U_0(\beta, \Gamma)$$

for all $\beta \in \mathcal{B}$. Furthermore, the equality holds when $\beta_t = 1$ for all $t \in [0, T]$ from (HJB_{PK}), i.e., u_0 is equal to $\tilde{U}_0(\Gamma)$. Thus, $\tilde{U}_0(\Gamma) \geq U_0(\beta, \Gamma)$ for all $\beta \in \mathcal{B}$, which implies incentive compatibility. $\square$

B.2.2 The Principal's Problem

We now explore how the principal's value function $V(u_t)$ evolves. Note that $V(0) = 0$ since the agent will not participate in the contract when the continuation utility is zero. This will serve as a boundary condition. The value function $V(u_t)$ can be heuristically written as follows:

$$V(u_t) = \max_{R_t, u_M^t, a_t} \begin{aligned} & - c dt + (\Pi - R_t) \lambda_D a_t dt + V_M(u_M^t) \lambda_S(1 - a_t) dt \\ & + \{1 - \lambda_D a_t dt - \lambda_S(1 - a_t) dt\} \cdot V(u_{t+dt}) \end{aligned}$$

subject to (IC).

By using $V(u_{t+dt}) = V(u_t) + V'(u_t)\dot{u}_t dt + o(dt)$, canceling $V(u_t)$ on both sides, taking the limit as $dt \rightarrow 0$ and plugging (HJB_{PK}) in, we obtain an HJB equation:

$$0 = \max_{R, u_M, a} \mathcal{J}(V(\cdot), R, u_M, a). \quad (\text{HJB}_V)$$

where

$$\begin{aligned} \mathcal{J}(V(\cdot), R, u_M, a) \equiv & -c + (\Pi - R - V(u))\lambda_D a + (V_M(u_M) - V(u))\lambda_S(1-a) \\ & - \{(R - u)\lambda_D a + (u_M - u)\lambda_S(1-a)\} \cdot V'(u) \end{aligned} \quad (\text{B.2})$$

Then, the principal's problem is to solve (HJB_V) subject to (IC) with the boundary condition $V(0) = 0$. The following lemma shows that the solution of the problem maximizes the principal's expected payoff subject to a promise keeping constraint $U_0(\Gamma) = u$.

Lemma 6 (Verification Lemma). *Suppose that a differentiable and concave function $\tilde{V}$ solves (HJB_V) subject to (IC) with the boundary condition $\tilde{V}(0) = 0$. Then, for any incentive-compatible contract Γ with $U_0(\Gamma) = u$,*

$$\tilde{V}(u) \geq \tilde{P}_0(\Gamma).$$

Proof of Lemma 6. Consider an arbitrary (deterministic) incentive-compatible contract Γ where the agent's expected payoff is given by u_t . The payoff to the principal under Γ is

$$\begin{aligned} \tilde{P}_0(\Gamma) &= \int_0^T (\Pi - R_t - c \cdot t) \cdot \lambda_D a_t \tilde{f}(a; 0, t) dt \\ &\quad + \int_0^T (V_M(u_M^t) - c \cdot t) \cdot \lambda_A (1 - a_t) \tilde{f}(a; 0, t) dt - c \cdot T \cdot \tilde{f}(a; 0, T) \\ &= \int_0^T ((\Pi - R_t) \cdot \lambda_D a_t + V_M(u_M^t) \cdot \lambda_A (1 - a_t) - c) \tilde{f}(a; 0, t) dt. \end{aligned}$$

Since $\tilde{V}$ solves the HJB equation, we have

$$0 \geq -c + (\Pi - R_t - \tilde{V}(u_t))\lambda_D a_t + (V_M(u_M^t) - \tilde{V}(u_t))\lambda_A(1 - a_t) \\ - \{(R_t - u_t) \cdot \lambda_D a_t + (u_M^t - u_t) \cdot \lambda_S(1 - a_t)\} \tilde{V}'(u_t).$$

By using (HJB_{PK}), rearranging, and multiplying by $\tilde{f}(a; 0, t)$, we can obtain that

$$(\lambda_D a_t + \lambda_A(1 - a_t))\tilde{f}(a; 0, t) \cdot \tilde{V}(u_t) - \tilde{f}(a; 0, t) \cdot \tilde{V}'(u_t)\dot{u}_t \\ \geq \tilde{f}(a; 0, t) \cdot ((\Pi - R_t) \cdot \lambda_D a_t + V_M(u_M^t) \cdot \lambda_A(1 - a_t) - c) \quad (\text{B.3})$$

Note that

$$\frac{d}{dt} \left(-\tilde{f}(a; 0, t) \cdot \tilde{V}(u_t) \right) = (\lambda_D a_t + \lambda_A(1 - a_t))\tilde{f}(a; 0, t) \cdot \tilde{V}(u_t) - \tilde{f}(a; 0, t) \cdot \tilde{V}'(u_t)\dot{u}_t.$$

Then, by integrating (B.3) over $[0, T]$ and noting that $\tilde{f}(a; 0, 0) = 1$, $u_T = 0$ and $\tilde{V}(0) = 0$, we have

$$\tilde{V}(u_0) = \tilde{V}(u_0) - \tilde{f}(a; 0, T)\tilde{V}(u_T) \\ \geq \int_0^T \tilde{f}(a; 0, t) \cdot ((\Pi - R_t) \cdot \lambda_D a_t + V_M(u_M^t) \cdot \lambda_A(1 - a_t) - c) dt = \tilde{P}_0(\Gamma).$$

Therefore, $\tilde{V}(u_0)$ is greater than or equal to any deterministic contract where the agent's expected payoff is equal to u_0 . Since $\tilde{V}$ is assumed to be concave, it is greater than or equal to any randomized contract. $\square$

B.3 Value Function Candidates and Implementation

Lemma 7. *The following statements hold.*

- (a) *An execution-only contract with the deadline u/ϕ implements a pair of expected payoffs*

of the principal and the agent $(V^B(u), u)$ where

$$V^B(u) \equiv \mathcal{W}^B\left(\frac{u}{\phi}\right) - u. \quad (\text{B.4})$$

(b) When $0 < u_1 < u$, a one-switch contract with the intermediate deadline $(u - u_1)/\phi$ and the final deadline u/ϕ implements $(V^{AB}(u|u_1), u)$ where

$$V^{AB}(u|u_1) \equiv \mathcal{W}_{u_1/\phi}^{AB}\left(\frac{u}{\phi}, \frac{1}{\lambda_S}\right) - u. \quad (\text{B.5})$$

(c) A training-only contract with the deadline u/ϕ implements $(V^{AB}(u|0), u)$.

(d) The following differential equations hold:

$$\phi V^{B'}(u) = \lambda_D \left(\Pi - \frac{\phi}{\lambda_D} - u - V^B(u) \right) - c, \quad (\text{B.6})$$

$$\phi V^{AB'}(u|u_1) = \lambda_S \left(V_M(u + \frac{\phi}{\lambda_S}) - V^{AB}(u|u_1) \right) - c. \quad (\text{B.7})$$

Together with Theorem 1, this lemma implies that $(\mathcal{V}(u), u)$ —defined in (4.2)—can be implemented by one of the above three contracts. Moreover, there exists $\hat{u}_1 \geq 0$ such that $\mathcal{V}$ can be rewritten as follows:

$$\mathcal{V}(u) = \begin{cases} V^B(u), & \text{if } u < \hat{u}_1, \\ V^{AB}(u|\hat{u}_1), & \text{if } u \geq \hat{u}_1. \end{cases} \quad (\text{B.8})$$

Specifically, $\hat{u}_1$ is chosen to be equal to $\phi \hat{T}$ if $1/\lambda_S < \bar{\Delta}$, and 0 if $1/\lambda_S \geq \bar{\Delta}$. The following lemma provides useful properties of $\mathcal{V}$ and $\hat{u}_1$.

Lemma 8. *Suppose that $\lambda_S = 2\lambda_D$. The following statements hold.*

(a) if $\hat{u}_1 > 0$, $V^{AB'}(\hat{u}_1|\hat{u}_1) = V^{B'}(\hat{u}_1)$ and $V^{AB'}(u|u) < V^{B'}(u)$ for all $u < u_1$, and if $\hat{u}_1 = 0$, $V^{AB'}(0|0) \geq V^{B'}(0)$.

(b) $\mathcal{V}'(u) \geq -1$ for all $u \geq 0$.

(c) $\mathcal{V}$ is concave.

B.3.1 Proof of Lemmas

Proof of Lemma 7. (a) Let $\Gamma_B(T)$ denote an execution-only contract with the deadline T .

The agent's expected payoff is

$$\begin{aligned} U_0(\Gamma_B(T)) &= \int_0^T R_\tau \lambda_D e^{-\lambda_D \tau} d\tau = \int_0^T \phi \left(T - \tau + \frac{1}{\lambda_D} \right) \lambda_D e^{-\lambda_D \tau} d\tau \\ &= -\phi(T - \tau) e^{-\lambda_D \tau} \Big|_0^T = \phi T. \end{aligned}$$

Therefore, $U_0(\Gamma_B(u/\phi)) = u$.

Also note that the sum of the expected payoffs of the principal and the agent should equal to the expected surplus from the execution-only policy with a deadline of T :

$$P_0(\Gamma_B(T)) + U_0(\Gamma_B(T)) = \mathcal{W}^B(T).$$

Therefore,

$$P_0(\Gamma_B(u/\phi)) = \mathcal{W}^B(u/\phi) - u = V^B(u).$$

(b) Let $\Gamma_{AB}(T_1, T)$ denote a contract with a switch from the training path to the execution path at T_1 and the deadline T . The subcontract at time $t \leq T_1$ is denoted by $\hat{\Gamma}_{AB}(t|T_1, T)$. Then, the agent's expected payoff for the subcontract $\hat{\Gamma}_{AB}(t|T_1, T)$ at

time t is

$$\begin{aligned} U_t(\hat{\Gamma}_{AB}(t|T_1, T)) &= \int_t^{T+\frac{1}{\lambda_S}} \phi\left(T + \frac{1}{\lambda_S} - \tau + \frac{1}{\lambda_S}\right) \lambda_S e^{-\lambda_S(\tau-t)} d\tau \\ &= -\phi\left(T + \frac{1}{\lambda_S} - \tau\right) e^{-\lambda_S(\tau-t)} \Big|_t^{T+\frac{1}{\lambda_S}} = \phi\left(T + \frac{1}{\lambda_S} - t\right). \end{aligned}$$

Also note that

$$\begin{aligned} \int_0^{T_1} U_\tau(\hat{\Gamma}_{AB}(\tau|T_1, T)) \lambda_S e^{-\lambda_S \tau} d\tau &= \int_0^{T_1} \phi\left(T + \frac{1}{\lambda_S} - \tau\right) \lambda_S e^{-\lambda_S \tau} d\tau \\ &= -\phi(T - \tau) e^{-\lambda_S \tau} \Big|_0^{T_1} = \phi T - \phi(T - T_1) e^{-\lambda_S T_1}. \end{aligned}$$

Then, the agent's expected payoff at time 0 is

$$\begin{aligned} U_0(\Gamma_{AB}(T_1, T)) &= \int_0^{T_1} U_\tau(\hat{\Gamma}_{AB}(\tau|T_1, T)) \lambda_S e^{-\lambda_S \tau} d\tau \\ &\quad + e^{-\lambda_S T_1} \int_{T_1}^T \phi\left(T + \frac{1}{\lambda_D} - \tau\right) \lambda_D e^{-\lambda_D(\tau-T_1)} d\tau \\ &= \phi T - \phi(T - T_1) e^{-\lambda_S T_1} - e^{-\lambda_S T_1} \left[\phi(T - \tau) e^{-\lambda_D(\tau-T_1)} \Big|_{T_1}^T \right] = \phi T. \end{aligned}$$

Thus, $U_0(\Gamma_{AB}(T_1, u/\phi)) = u$.

As in the previous case, the sum of the expected payoffs of the principal and the agent is equal to the one-switch policy with the intermediate deadline T_1 , the deadline T , and the extension $1/\lambda_S$:

$$P_0(\Gamma_{AB}(T_1, T)) + U_0(\Gamma_{AB}(T_1, T)) = \mathcal{W}_{T-T_1}^{AB}(T, 1/\lambda_S).$$

By plugging in $T = u/\phi$ and $T_1 = (u - u_1)/\phi$, (B.5) holds.

- (c) Note that a training-only contract with a deadline T is equivalent to a contract with a switch from the training path to the execution path at $T_1 = T$ and a deadline

T . Therefore, by the previous result, a training-only contract with the deadline u/ϕ implements $(V^{AB}(u|0), u)$.

(d) By the construction of $\mathcal{W}^B$,

$$\dot{\mathcal{W}}^B(T) = \lambda_D(\Pi - \mathcal{W}^B(T)) - c, \quad (\text{B.9})$$

for all $T \geq 0$. Similarly,

$$\dot{\mathcal{W}}_{\hat{T}}^{AB}(T, 1/\lambda_S) = \lambda_S(W_{T+1/\lambda_S}^1 - \mathcal{W}_{\hat{T}}^{AB}(T, 1/\lambda_S)) - c, \quad (\text{B.10})$$

for all $T \geq \hat{T}$.

Using the definitions of V^B , V^{AB} and V_M , (B.6) and (B.7) follow.

□

Proof of Lemma 8. (a) Suppose that $\hat{u}_1 > 0$, which implies that $1/\lambda_S < \bar{\Delta}$. Now set $\Delta = 1/\lambda_S$. In Lemma ??, $\hat{T}$ is chosen to satisfy $\lambda_D(\Pi - \mathcal{W}^B(\hat{T})) = \lambda_S(W_{\hat{T}+1/\lambda_S}^1 - \mathcal{W}^B(\hat{T}))$ and $\lambda_D(\Pi - \mathcal{W}^B(z)) > \lambda_S(W_{z+1/\lambda_S}^1 - \mathcal{W}^B(z))$ for all $z < \hat{T}$.

Using Lemma 7 (d) and $\mathcal{W}^B(\hat{T}) = \mathcal{W}_{\hat{T}}^{AB}(\hat{T}, 1/\lambda_S)$, we can derive that $V^{AB'}(\hat{u}_1|\hat{u}_1) = V^{B'}(\hat{u}_1)$ and $V^{AB'}(u|u) < V^{B'}(u)$ for all $u < \hat{u}_1$.

When $\hat{u}_1 = 0$, and thereby $1/\lambda_S \geq \bar{\Delta}$, it follows from (A.10) that $V^{AB'}(\hat{u}_1|\hat{u}_1) \geq V^{B'}(\hat{u}_1)$.

(b) Since a longer the deadline increases the expected social surplus, $\mathcal{W}^*(T, \Delta)$ is increasing in T . Therefore,

$$\mathcal{V}'(u) = \frac{\dot{\mathcal{W}}^*(u/\phi, 1/\lambda_S)}{\phi} - 1 \geq -1.$$

(c) If $u \leq \hat{u}_1$,

$$\mathcal{V}''(u) = V^{B''}(u) = -\left(\Pi - \frac{c}{\lambda_D}\right) \frac{\lambda_D^2}{\phi^2} e^{-\frac{\lambda_D}{\phi}u} < 0.$$

Now assume that $u > \hat{u}_1$. By using (B.9), (B.10) and $V^{B'}(\hat{u}_1) \leq V^{AB'}(\hat{u}_1|\hat{u}_1)$, we can derive that

$$\mathcal{W}^B(\hat{u}_1/\phi) \leq \Pi - \frac{c}{\lambda_S - \lambda_D} - \frac{\lambda_S}{\lambda_S - \lambda_D} \left(\Pi - \frac{c}{\lambda_S} \right) e^{-\frac{\lambda_S}{\phi} \hat{u}_1 - 1}. \quad (\text{B.11})$$

Using (??), we can derive the followings though some algebra:

$$\begin{aligned} V^{AB''}(u|\hat{u}_1) &= \left(\frac{\lambda_S}{\phi} \right)^2 e^{\frac{\lambda_S}{\phi}(\hat{u}_1 - u)} \left[\mathcal{W}^B(\hat{u}_1/\phi) - \left(\Pi - \frac{2c}{\lambda_S} \right) + 2 \left(\Pi - \frac{c}{\lambda_S} \right) e^{-\frac{\lambda_S}{\phi} \hat{u}_1 - 1} \right] \\ &\quad - \left(\Pi - \frac{c}{\lambda_S} \right) \left(\frac{\lambda_S}{\phi} \right)^3 (u - \hat{u}_1) e^{-\frac{\lambda_S}{\phi}(u + \frac{1}{\lambda_S})} \end{aligned}$$

By plugging (B.11) in, we have

$$\begin{aligned} V^{AB''}(u|\hat{u}_1) &\leq \left(\frac{\lambda_S}{\phi} \right)^2 e^{\frac{\lambda_S}{\phi}(\hat{u}_1 - u)} \left[\frac{\lambda_S - 2\lambda_D}{\lambda_S(\lambda_S - \lambda_D)} c + \frac{\lambda_S - 2\lambda_D}{\lambda_S - \lambda_D} \left(\Pi - \frac{c}{\lambda_S} \right) e^{-\frac{\lambda_S}{\phi} \hat{u}_1 - 1} \right] \\ &\quad - \left(\Pi - \frac{c}{\lambda_S} \right) \left(\frac{\lambda_S}{\phi} \right)^3 (u - \hat{u}_1) e^{-\frac{\lambda_S}{\phi}(u + \frac{1}{\lambda_S})}. \end{aligned} \quad (\text{B.12})$$

Then, from $\lambda_S = 2\lambda_D$, $\mathcal{V}''(u) = V^{AB''}(u|\hat{u}_1) \leq 0$ for all $u \geq \hat{u}_1$.

□

B.4 Value Function Verification (Proposition 2)

The goal of this subsection is to prove Proposition 2. Specifically, we show that the value function defined in the previous section solves (HJB_V) subject to (IC). To achieve this, we introduce functions that specify potential deviations and then establish useful properties as a lemma, followed by the proof for Proposition 2.

First, define

$$L^B(u, R) \equiv \mathcal{J}(\mathcal{V}(\cdot), R, \cdot, 1) = \lambda_D(\Pi - R - \mathcal{V}(u)) - c - \lambda_D(R - u)\mathcal{V}'(u). \quad (\text{B.13})$$

Given u , maximizing this function with respect to $R \geq u + \phi/\lambda_D$ is equivalent to maximizing the right hand side of (HJB_V) under (IC) with $a = 1$.

Similarly, define

$$L^A(u, w) \equiv \mathcal{J}(\mathcal{V}(\cdot), \cdot, w, 0) = \lambda_S(V_M(w) - \mathcal{V}(u)) - c - \lambda_S(w - u)\mathcal{V}'(u). \quad (\text{B.14})$$

Given u , maximizing this function with respect to $w \geq u + \phi/\lambda_S$ is equivalent to maximizing the right hand side of (HJB_V) under (IC) with $a = 0$.

Lemma 9. *Suppose that $\Pi > c/\lambda_D$ and $\lambda_S = 2\lambda_D$. Then, for any $u \geq 0$, $L^B(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_D$, and $L^A(u, w) \leq 0$ for all $w \geq u + \phi/\lambda_S$.*

Proof of Lemma 9. We begin by showing $L^B(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_D$. Observe that

$$\frac{\partial L^B}{\partial R} = -\lambda_D(1 + \mathcal{V}'(u)) \leq 0.$$

from Lemma 8 (b). Also note that

$$\begin{aligned} L^B(u, u + \phi/\lambda_D) &= \lambda_D(\Pi - u - \mathcal{V}(u)) - c - \phi(\mathcal{V}'(u) + 1) \\ &= \lambda_D(\Pi - \mathcal{W}^*(u/\phi, 1/\lambda_S)) - c - \dot{\mathcal{W}}^*(u/\phi, 1/\lambda_S) \leq 0. \end{aligned}$$

The last inequality is due to (HJB_W). Therefore, $L^B(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_D$.

For L^A , observe that

$$\frac{\partial L^A}{\partial u} = -\lambda_S(w - u)\mathcal{V}''(u) \geq 0$$

by the concavity of $\mathcal{V}$. Therefore, it is sufficient to check whether $L^A(u, u + \phi/\lambda_S) \leq 0$ for all $u \geq 0$.

Note that for all $u \geq \hat{u}_1$, $L^A(u, u + \phi/\lambda_S) = 0$ holds by (B.7). Now, suppose that $u < \hat{u}_1$, thereby $\mathcal{V}(u) = V^B(u)$. Using (B.7) and Lemma 8 (a), we have $L^A(u, u + \phi/\lambda_S) = \phi(V^{AB'}(u|u) - V^{B'}(u)) < 0$ for all $u < \hat{u}_1$. $\square$

Now We prove Proposition 2.

Proof of Proposition 2. We begin by showing that $\mathcal{J}$ becomes zero when the value function $\mathcal{V}$ is utilized alongside contractual terms with binding ICs.

When $u \geq \hat{u}_1$, $\mathcal{V}(u) = V^{AB}(u|\hat{u}_1)$. Then, by (B.7),

$$\mathcal{J}(V^{AB}(u|\hat{u}_1), \cdot, u + \phi/\lambda_S, 0) = 0.$$

Likewise, when $u < \hat{u}_1$, $\mathcal{V}(u) = V^B(u)$, and by (B.6),

$$\mathcal{J}(V^B(u), u + \phi/\lambda_D, \cdot, 1) = 0.$$

Lemma 9 shows that $\mathcal{J}$ is nonpositive for any feasible deviations. Therefore, $\mathcal{V}$ solves (HJB $_{\mathcal{V}}$) subject to (IC).

The concavity of $\mathcal{V}$ is shown in Lemma 8 (c). If $\hat{u}_1 = 0$, $\mathcal{V}(u) = V^{AB}(u|0)$ is differentiable for all $u \geq 0$. If $\hat{u}_1 > 0$, $V^B(u)$ is differentiable for all $u < \hat{u}_1$ and $V^{AB}(u|\hat{u}_1)$ is differentiable for all $u > \hat{u}_1$. By Lemma 8 (a), $\mathcal{V}$ is differentiable at $\hat{u}_1$ as well. Also note that $\mathcal{V}(0) = 0$. Therefore, by Lemma 6, for any incentive compatible contract promising the agent u units of utility, the principal's expected payoff is lower than or equal to $\mathcal{V}(u)$.

Last, by Lemma 7, there exists a contract implementing $(\mathcal{V}(u), u)$. Therefore, $\mathcal{V}(u)$ is the principal's maximized expected payoff subject to the promise-keeping constraint $U_0(\Gamma) = u$ and the incentive compatibility constraints. $\square$

B.5 Thresholds

In this section, we explain how to pin down the thresholds π_B and π_A and provide some properties of them.

First, recall that π_A is the solution of $1/\lambda_S = \bar{\Delta}$. This gives us

$$\frac{1}{\lambda_S} = \frac{1}{\lambda_S} \log \left[\frac{2\pi - 1}{\pi - 1} \right] \quad \Leftrightarrow \quad \pi_A = \frac{e - 1}{e - 2} \approx 2.392. \quad (\text{B.15})$$

Also recall that the threshold is relevant to whether the switching point $\hat{u}_1$ is greater than $\bar{u}$ or not. Since $\mathcal{V}$ is concave, $\hat{u}_1 \leq \bar{u}$ if and only if $\mathcal{V}'(\hat{u}_1) \geq 0$. Observe that by using the formula of V^B and (A.13), we can derive that

$$\mathcal{V}'(\hat{u}_1) = V^{B'}(\hat{u}_1) = \frac{(\lambda_D \Pi - c)^2}{\phi \cdot (\lambda_S \Pi - c)} e - 1.$$

By solving the equation making the above equal to zero, it follows that $\mathcal{V}'(\hat{u}_1) \geq 0$ if and only if

$$\pi \leq \pi_B \equiv \frac{c \cdot e + \phi + \sqrt{\phi(c \cdot e + \phi)}}{c \cdot e}. \quad (\text{B.16})$$

We conclude the section by showing that π_B lies between π_F and π_A .

Lemma 10. *When $\lambda_S = 2\lambda_D$, $\pi_B \in (\pi_F, \pi_A)$.*

Proof of Lemma 10. From $\phi \leq c$, we have

$$\pi_B \leq \frac{(e + 1) + \sqrt{e + 1}}{e} \approx 2.077.$$

Therefore, by (B.15), $\pi_B < \pi_A$.

Next, observe that

$$\pi_B - \pi_F = \frac{-\phi(e - 1) + \sqrt{\phi(c \cdot e + \phi)}}{c \cdot e} \geq \frac{\phi(\sqrt{e + 1} - (e - 1))}{c \cdot e}.$$

Since $\sqrt{e + 1} - (e - 1) \approx .21 > 0$, $\pi_B > \pi_F$. □

B.6 Proof of Proposition 3

Throughout this subsection we assume $\lambda_D < \lambda_S < 2\lambda_D$, i.e. $0 < \eta < 1$. For a candidate surplus function w it is convenient to abbreviate the gap defined in (A.6) as

$$\Lambda_z[w] \equiv \Lambda_z[w] = \lambda_S (W_{z+\Delta}^1 - w(z)) - \lambda_D (\Pi - w(z)), \quad (\text{B.17})$$

the difference between the two branches of (HJB_W): the staged route attains the maximum in (HJB_W) at z if and only if $\Lambda_z[w] \geq 0$. The two substituted forms used in the proof of Proposition 1 are $\Lambda_1(x)$ of (A.11), obtained by setting $w = \mathcal{W}^D$ and $x = e^{-\lambda_D z}$, and $\Lambda_2(y)$ of (A.14), obtained by setting $w = \mathcal{W}_Z^{SD}$ and $y = e^{-\lambda_S(z-Z)}$.

The proof of Proposition 1 draws its conclusions from $\lambda_S = 2\lambda_D$ in exactly two places, and they are the same place: the two substituted forms satisfy

$$\lim_{x \rightarrow 0} \Lambda_1(x) = \frac{\lambda_S - 2\lambda_D}{\lambda_D} \cdot c \quad \text{and} \quad \lim_{y \rightarrow 0} \Lambda_2(y) = \frac{\lambda_S - 2\lambda_D}{\lambda_S} \cdot c, \quad (\text{B.18})$$

and both vanish at $\lambda_S = 2\lambda_D$, which is what allows each of Lemmas 3 and 4 to conclude that a sign holds on an entire half-line.¹⁷ When $\lambda_S < 2\lambda_D$ both limits are strictly negative, and each conclusion weakens in the same way, from a sign to a single crossing. The following lemma converts these two single crossings into a policy.

Lemma 11 (Crossing rate). *Let w be differentiable on an interval $I \subseteq \mathbb{R}_+$ and satisfy either (A.7) or (A.8) on I . Then at every $z \in I$ with $\Lambda_z[w] = 0$,*

$$\frac{d}{dz} \Lambda_z[w] = \Phi(z) \equiv \lambda_S(\lambda_S - \lambda_D) (\Pi - W_{z+\Delta}^1) - \lambda_D c, \quad (\text{B.19})$$

irrespective of which of the two equations w satisfies. Moreover Φ is strictly decreasing, with

$$\lim_{z \rightarrow \infty} \Phi(z) = (\lambda_S - 2\lambda_D)c < 0.$$

¹⁷The knife-edge is also what reduces Λ_1 to the quadratic (A.12) and hence delivers $\hat{x}$ in closed form. Under an efficiency loss we retain the general form (A.11) and characterize the thresholds implicitly.

Proof of Lemma 11. From (A.5), $\frac{d}{dz}W_{z+\Delta}^1 = \lambda_S(\Pi - \frac{c}{\lambda_S} - W_{z+\Delta}^1)$, so

$$\frac{d}{dz}\Lambda_z[w] = \lambda_S^2 \left(\Pi - \frac{c}{\lambda_S} - W_{z+\Delta}^1 \right) - (\lambda_S - \lambda_D)\dot{w}(z).$$

At a point with $\Lambda_z[w] = 0$ we have $(\lambda_S - \lambda_D)w(z) = \lambda_S W_{z+\Delta}^1 - \lambda_D \Pi$, and therefore

$$\begin{aligned} (\lambda_S - \lambda_D)(\Pi - w(z)) &= \lambda_S(\Pi - W_{z+\Delta}^1), \\ (\lambda_S - \lambda_D)(W_{z+\Delta}^1 - w(z)) &= \lambda_D(\Pi - W_{z+\Delta}^1). \end{aligned}$$

Substituting the first identity into (A.7) and the second into (A.8) gives, in both cases,

$$(\lambda_S - \lambda_D)\dot{w}(z) = \lambda_S \lambda_D (\Pi - W_{z+\Delta}^1) - (\lambda_S - \lambda_D)c,$$

and (B.19) follows. Since $W_{z+\Delta}^1$ is strictly increasing in z and converges to $\Pi - c/\lambda_S$, the function Φ is strictly decreasing with limit $\lambda_S(\lambda_S - \lambda_D)\frac{c}{\lambda_S} - \lambda_D c = (\lambda_S - 2\lambda_D)c$, which is negative because $\lambda_S < 2\lambda_D$. $\square$

Corollary 12 (No return). *Suppose w satisfies (A.7) on $[z_0, \infty)$, with $\Lambda_{z_0}[w] = 0$ and $\Phi(z_0) < 0$. Then $\Lambda_z[w] < 0$ for all $z > z_0$.*

Proof of Corollary 12. Since Φ is strictly decreasing, $\Phi(z) < 0$ for all $z \geq z_0$. By Lemma 11 the derivative of $\Lambda_z[w]$ at z_0 equals $\Phi(z_0) < 0$, so $\Lambda_z[w] < 0$ for z slightly above z_0 . Suppose $\Lambda_z[w] = 0$ for some $z > z_0$ and let z_1 be the smallest such point. Then $\Lambda_z[w] < 0$ on (z_0, z_1) and $\Lambda_{z_1}[w] = 0$, so the derivative at z_1 is non-negative; but Lemma 11 gives $\Phi(z_1) < 0$, a contradiction. $\square$

The next two lemmas are Lemmas 3 and 4 with $\lambda_S = 2\lambda_D$ relaxed. Neither of the two concavity properties they rely on uses the knife-edge restriction.

Lemma 13. *Suppose that $\lambda_D < \lambda_S < 2\lambda_D$ and $\Delta \leq \bar{\Delta}$. Then Λ_1 is strictly concave on $(0, 1]$ and attains its maximum at*

$$x^* \equiv \left[\frac{\lambda_D(\lambda_S - \lambda_D) \left(\Pi - \frac{c}{\lambda_D} \right)}{\lambda_S(\lambda_S \Pi - c)} \cdot e^{\lambda_S \Delta} \right]^{\frac{\lambda_D}{\lambda_S - \lambda_D}}. \quad (\text{B.20})$$

Moreover:

(i) if $x^* \geq 1$ or $\Lambda_1(x^*) \leq 0$, then $\Lambda_z[\mathcal{W}^D] \leq 0$ for all $z \geq 0$, with equality at most at one point;

(ii) otherwise there is a unique $\hat{x} \in (x^*, 1]$ with $\Lambda_1(\hat{x}) = 0$, and, setting $\hat{T} \equiv -\log(\hat{x})/\lambda_D \geq 0$, we have $\Lambda_z[\mathcal{W}^D] < 0$ for all $z < \hat{T}$, $\Lambda_{\hat{T}}[\mathcal{W}^D] = 0$, and $\Phi(\hat{T}) > 0$.

Proof of Lemma 13. Differentiating (A.11) twice,

$$\Lambda_1''(x) = -(\lambda_S \Pi - c) e^{-\lambda_S \Delta} \frac{\lambda_S}{\lambda_D} \left(\frac{\lambda_S}{\lambda_D} - 1 \right) x^{\frac{\lambda_S}{\lambda_D} - 2} < 0,$$

where the sign uses $\lambda_S > \lambda_D$ and $\lambda_S \Pi > \lambda_D \Pi > c$ from (2.1). Setting $\Lambda_1'(x) = 0$ and solving gives (B.20); since $\lambda_S > \lambda_D$ we have $\Lambda_1'(x) \rightarrow (\lambda_S - \lambda_D)(\Pi - \frac{c}{\lambda_D}) > 0$ as $x \rightarrow 0$, so $x^* > 0$ is the global maximizer. Evaluating (A.11) at the two ends of the domain,

$$\lim_{x \rightarrow 0} \Lambda_1(x) = \frac{\lambda_S - 2\lambda_D}{\lambda_D} c < 0 \quad \text{and} \quad \Lambda_1(1) = (\lambda_S \Pi - c) (1 - e^{-\lambda_S \Delta}) - \lambda_D \Pi,$$

and $\Lambda_1(1) \leq 0$ is precisely (A.10) reversed, i.e. $\Delta \leq \bar{\Delta}$ with $\bar{\Delta}$ as in (5.2).

If $x^* \geq 1$ then Λ_1 is strictly increasing on $(0, 1)$, so $\Lambda_1(x) < \Lambda_1(1) \leq 0$ there; if instead $\Lambda_1(x^*) \leq 0$ then $\Lambda_1(x) \leq \Lambda_1(x^*) \leq 0$ for all x , with equality only possible at x^* . In either case $\Lambda_z[\mathcal{W}^D] \leq 0$ for all z , which is (i).

Otherwise $x^* < 1$ and $\Lambda_1(x^*) > 0 \geq \Lambda_1(1)$. Since Λ_1 is strictly decreasing on $[x^*, 1]$, it has a unique zero $\hat{x}$ there, and $\hat{x} > x^*$. Strict concavity and $\Lambda_1'(x^*) = 0$ give $\Lambda_1'(\hat{x}) < 0$,

and $\Lambda_1(x) < 0$ for all $x \in (\hat{x}, 1]$. Because $x = e^{-\lambda_D z}$ is strictly decreasing in z , this says $\Lambda_z[\mathcal{W}^D] < 0$ for all $z < \hat{T}$, with equality at $\hat{T}$. Finally $\mathcal{W}^D$ satisfies (A.7) on $\mathbb{R}_+$, so Lemma 11 applies at $\hat{T}$ and the chain rule gives

$$\Phi(\hat{T}) = \frac{d}{dz} \Lambda_z[\mathcal{W}^D] \Big|_{z=\hat{T}} = -\lambda_D \hat{x} \Lambda'_1(\hat{x}) > 0. \quad \square$$

Lemma 14. *Suppose that $\lambda_D < \lambda_S < 2\lambda_D$ and that $Z \geq 0$ satisfies either $\Lambda_Z[\mathcal{W}^D] > 0$, or $\Lambda_Z[\mathcal{W}^D] = 0$ and $\Phi(Z) > 0$. Then Λ_2 is strictly concave on $(0, 1]$ and there is a unique $\check{y} \in (0, 1)$ with $\Lambda_2(\check{y}) = 0$. Setting $\check{T} \equiv Z - \log(\check{y})/\lambda_S > Z$, we have $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for all $z \in (Z, \check{T})$, $\Lambda_{\check{T}}[\mathcal{W}_Z^{SD}] = 0$, and $\Phi(\check{T}) < 0$.*

Proof of Lemma 14. Strict concavity is $\Lambda_2''(y) = -(\lambda_S - \lambda_D)\kappa/y < 0$, and the limit in (B.18) is $\Lambda_Z[\mathcal{W}^D] + h = \frac{\lambda_S - 2\lambda_D}{\lambda_S} c$, which is strictly negative because $\lambda_S < 2\lambda_D$. Neither computation uses the knife-edge restriction.

We first show that $\Lambda_2(y) > 0$ for y slightly below 1. If $\Lambda_Z[\mathcal{W}^D] > 0$ this is immediate from $\Lambda_2(1) = \Lambda_Z[\mathcal{W}^D]$ and continuity. If instead $\Lambda_Z[\mathcal{W}^D] = 0$, then $\mathcal{W}_Z^{SD}$ satisfies (A.8) on $[Z, \infty)$, so Lemma 11 gives a derivative of $\Phi(Z) > 0$ at Z and hence $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for z slightly above Z , which is the same statement since $y = e^{-\lambda_S(z-Z)}$ is strictly decreasing in z .

Because Λ_2 is concave, the superlevel set $\{y \in (0, 1] : \Lambda_2(y) \geq 0\}$ is an interval. It contains a left neighborhood of 1 and, by the limit above, excludes a right neighborhood of 0; hence it equals $[\check{y}, 1]$ for a unique $\check{y} \in (0, 1)$, with $\Lambda_2 > 0$ on $(\check{y}, 1)$ and $\Lambda_2 < 0$ on $(0, \check{y})$. Translating back, $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for $z \in (Z, \check{T})$ and vanishes at $\check{T}$.

Finally, Λ_2 is positive somewhere on $(\check{y}, 1)$, so $\check{y}$ lies strictly below the maximizer of Λ_2 and strict concavity gives $\Lambda'_2(\check{y}) > 0$. Applying Lemma 11 at $\check{T}$ and the chain rule,

$$\Phi(\check{T}) = \frac{d}{dz} \Lambda_z[\mathcal{W}_Z^{SD}] \Big|_{z=\check{T}} = -\lambda_S \check{y} \Lambda'_2(\check{y}) < 0. \quad \square$$

It remains to extend the candidate beyond $\check{T}$. Let $\mathcal{W}_{Z_1, Z_2}^{SD}(z, \Delta)$ denote the expected

surplus of the two-switch policy that uses the direct route for $z < Z_1$, the staged route for $Z_1 \leq z \leq Z_2$, and the direct route again for $z > Z_2$. Solving (A.7) on (Z_2, ∞) with the value $\mathcal{W}_{Z_1}^{SD}(Z_2, \Delta)$ at Z_2 ,

$$\mathcal{W}_{Z_1, Z_2}^{DSD}(z, \Delta) = \begin{cases} \mathcal{W}_{Z_1}^{SD}(z, \Delta), & \text{if } z \leq Z_2, \\ \left(\Pi - \frac{c}{\lambda_D} \right) - \left[\left(\Pi - \frac{c}{\lambda_D} \right) - \mathcal{W}_{Z_1}^{SD}(Z_2, \Delta) \right] e^{-\lambda_D(z-Z_2)}, & \text{if } z > Z_2. \end{cases} \quad (\text{B.21})$$

Lemma 15. *Under the hypotheses of Lemma 14, $\Lambda_z[\mathcal{W}_{Z, \check{T}}^{DSD}] < 0$ for all $z > \check{T}$.*

Proof of Lemma 15. By (B.21) the function $\mathcal{W}_{Z, \check{T}}^{DSD}$ agrees with $\mathcal{W}_Z^{SD}$ at $\check{T}$, so $\Lambda_{\check{T}}[\mathcal{W}_{Z, \check{T}}^{DSD}] = 0$ by Lemma 14, and it satisfies (A.7) on $[\check{T}, \infty)$. That same lemma gives $\Phi(\check{T}) < 0$, so Corollary 12 applies. $\square$

Proof of Proposition 3. We construct $\hat{T}$ and $\check{T}$ and verify that $\mathcal{W}_{\hat{T}, \check{T}}^{DSD}$ solves (HJB_W) on $\mathbb{R}_+$ with value 0 at $z = 0$. Neither threshold depends on T , because (HJB_W) is stated in remaining time and its boundary condition is imposed at $z = 0$.

- (a) Suppose $\Delta > \bar{\Delta}$. Set $\hat{T} = 0$. Since $\mathcal{W}^D(0) = 0$, we have $\Lambda_0[\mathcal{W}^D] = \lambda_S W_\Delta^1 - \lambda_D \Pi > 0$ by (A.10), so Lemma 14 applies at $Z = 0$ and delivers $\check{T} \in (0, \infty)$.
- (b) Suppose $\Delta \leq \bar{\Delta}$, and apply Lemma 13. In case (i) put $\hat{T} = \check{T} = \infty$: then $\Lambda_z[\mathcal{W}^D] \leq 0$ everywhere, so $\mathcal{W}^D$ solves (HJB_W) on $\mathbb{R}_+$ by (A.7), and the direct-only policy is optimal for every T . In case (ii) we obtain $\hat{T} \geq 0$ with $\Lambda_{\hat{T}}[\mathcal{W}^D] = 0$ and $\Phi(\hat{T}) > 0$, which is the second hypothesis of Lemma 14 at $Z = \hat{T}$; that lemma then delivers $\check{T} \in (\hat{T}, \infty)$.

In either case, write $w \equiv \mathcal{W}_{\hat{T}, \check{T}}^{DSD}(\cdot, \Delta)$. On $[0, \hat{T})$, $w = \mathcal{W}^D$ satisfies (A.7) and $\Lambda_z[w] < 0$ by Lemma 13, so the direct branch attains the maximum in (HJB_W). On $(\hat{T}, \check{T})$, $w = \mathcal{W}_{\hat{T}}^{SD}$ satisfies (A.8) and $\Lambda_z[w] > 0$ by Lemma 14, so the staged branch attains it. On $(\check{T}, \infty)$, w

satisfies (A.7) and $\Lambda_z[w] < 0$ by Lemma 15, so the direct branch attains it again. At $\hat{T}$ and at $\check{T}$ we have $\Lambda_z[w] = 0$, so the two branches of (HJB_w) have equal right-hand sides there; consequently the two one-sided derivatives of w agree and w is differentiable at both points. Since $w(0) = \mathcal{W}^D(0) = 0$, the verification argument following (HJB_w) gives $w(z) = W_z^{0,\Delta}$ for all $z \geq 0$.

Fix a deadline T and translate remaining time into calendar time through $t = T - z$. The optimal policy uses the direct route when $z < \hat{T}$ or $z > \check{T}$ and the staged route when $z \in (\hat{T}, \check{T})$, which is the two-switch policy of Definition 3 with $S_1 = \max\{T - \check{T}, 0\}$ and $S_2 = \max\{T - \hat{T}, 0\}$. Reading off the degenerate cases gives the statements in (a) and (b): the policy is direct-only when $T \leq \hat{T}$, is a one-switch policy when $\hat{T} < T \leq \check{T}$, and has both switches interior when $T > \check{T}$.

Finally, dividing (A.11) by c and substituting $\eta = \lambda_S/\lambda_D - 1$ and $\pi = \lambda_D\Pi/c$ turns (B.20) into (OA.1.5) and $\Lambda_1(x^*)/c$ into $L(\eta, \pi, \Delta)$ of (5.4), so the dichotomy in Lemma 13 is the one stated in part (b) of the proposition. $\square$

Remark 3. As $\eta \uparrow 1$ the limit $\lim_{y \rightarrow 0} \Lambda_2(y) = \frac{\lambda_S - 2\lambda_D}{\lambda_S} c$ increases to zero. Since Λ_2 is concave and remains strictly positive on a fixed interior interval, its lower zero $\check{y}$ must then converge to zero, so $\check{T} = Z - \log(\check{y})/\lambda_S \uparrow \infty$: the outer direct phase is pushed beyond every finite deadline and Proposition 3 converges to Proposition 1.

Online Appendix for “Execution vs. Training under Endogenous Deadlines”

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In this online appendix, we provide the proofs for Section 5, specifically the value function and optimal contract characterizations when there is an efficiency loss from training (Proposition 6, Proposition 4, and Proposition 5). Some results in Appendix B can still be utilized (e.g., Lemma 5, Lemma 6, and Lemma 7) as they do not use the parametric assumption $\lambda_A = 2\lambda_B$.

OA.1 Value Function Characterization

We begin by specifying a value function that can be implemented by a two-switch contract defined in Definition 4.

Lemma OA.1. *The following statements hold.*

- (a) *When $0 < u_1 < u_2 < u$, a two-switch contract with the intermediate deadlines $(u - u_2)/\phi$, $(u - u_1)/\phi$ and the final deadline u/ϕ implements $(V^{BAB}(u|u_1, u_2), u)$ where*

$$V^{BAB}(u|u_1, u_2) \equiv \left(\Pi - \frac{c}{\lambda_B} \right) \left(1 - e^{\frac{\lambda_B}{\phi}(u_2 - u)} \right) + (V^{AB}(u_2|u_1) + u_2) e^{\frac{\lambda_B}{\phi}(u_2 - u)} - u. \quad (\text{OA.1.1})$$

- (b) *The following differential equation holds:*

$$\phi V^{BAB'}(u|u_1, u_2) = \lambda_B \left(\Pi - \frac{\phi}{\lambda_B} - u - V^{BAB}(u|u_1, u_2) \right) - c. \quad (\text{OA.1.2})$$

Proof of Lemma OA.1. (a) Let $\Gamma_{BAB}(T_1, T_2, T)$ denote a contract with two switches at $T_1 = (u - u_2)/\phi$ and $T_2 = (u - u_1)/\phi$ and a deadline $T = u/\phi$. Note that at time T_1 (if the problem has not been solved), the remaining contract is equivalent to $\Gamma_{AB}(T_2 - T_1, T - T_1)$. Recall that $U_0(\Gamma_{AB}(T_2 - T_1, T - T_1)) = \phi(T - T_1)$. Then, the agent's expected payoff at time 0 is

$$\begin{aligned} U_0(\Gamma_{BAB}(T_1, T_2, T)) &= \int_0^{T_1} \phi(T + 1/\lambda_B - \tau) \lambda_B e^{-\lambda_B \tau} d\tau \\ &\quad + e^{-\lambda_A T_1} U_0(\Gamma_{AB}(T_2 - T_1, T - T_1)) \\ &= \phi T - \phi(T - T_1) e^{-\lambda_B T_1} + e^{-\lambda_B T_1} \phi(T - T_1) = \phi T = u. \end{aligned}$$

Also note that

$$\begin{aligned} &P_0(\Gamma_{BAB}(T_1, T_2, T)) + U_0(\Gamma_{BAB}(T_1, T_2, T)) \\ &= \int_0^{T_1} (\Pi - c\tau) \lambda_B e^{-\lambda_B \tau} d\tau - cT_1 e^{-\lambda_B T_1} \\ &\quad + e^{-\lambda_B T_1} (P_0(\Gamma_{AB}(T_2 - T_1, T - T_1)) + U_0(\Gamma_{AB}(T_2 - T_1, T - T_1))). \end{aligned}$$

Recall that $U_0(\Gamma_{AB}(T'_2, T'_1)) + P_0(\Gamma_{AB}(T'_2, T'_1)) = V^{AB}(\phi T'_1 | \phi(T'_1 - T'_2)) + \phi(T'_1 - T'_2)$.

By plugging in $T'_1 = T - T_1$, $T'_2 = T_2 - T_1$, $\phi T_1 = u - u_2$ and $\phi T_2 = u - u_1$, the right hand side of the above equation is equal to:

$$\left(\Pi - \frac{c}{\lambda_B} \right) (1 - e^{-\lambda_B T_1}) + e^{-\lambda_B T_1} \cdot \{V^{AB}(u_2 | u_1) + u_2\} = V^{BAB}(u) + u,$$

thus, $P_0(\Gamma_{BAB}(T_1, T_2, T)) = V^{BAB}(u | u_1, u_2)$.

(b) Last, by taking the derivative of (OA.1.1) and multiplying by ϕ , we have

$$\begin{aligned}\phi V^{BAB'}(u|u_1, u_2) &= \lambda_B \left(\Pi - \frac{c}{\lambda_B} \right) e^{\frac{\lambda_B}{\phi}(u_2-u)} - \lambda_B (V^{AB}(u_2|u_1) + u_2) e^{\frac{\lambda_B}{\phi}(u_2-u)} - \phi \\ &= \lambda_B \left(\Pi - \frac{\phi}{\lambda_B} - u - V^{BAB}(u|u_1, u_2) \right) - c,\end{aligned}$$

thus, (OA.1.2) holds. □

Based on the intuition presented in the main text, we conjecture the value function defined as follows.

$$\mathcal{V}(u) = \begin{cases} V^B(u), & \text{if } 0 \leq u \leq \hat{u}_1, \\ V^{AB}(u|\hat{u}_1), & \text{if } \hat{u}_1 < u \leq \hat{u}_2, \\ V^{BAB}(u|\hat{u}_1, \hat{u}_2), & \text{if } \hat{u}_2 < u. \end{cases} \quad (\text{OA.1.3})$$

The following proposition shows that there exist $\hat{u}_1$ and $\hat{u}_2$ such that the above three value functions are smoothly pasted, and the resulting function is the principal's value function.

Proposition 6. *Suppose that η is less than 1.*

(a) (**Smooth Pasting**) *There exist $\hat{u}_2 \geq \hat{u}_1 \geq 0$ such that*

- i. $V^{B'}(u) > V^{AB'}(u|u)$ for all $0 \leq u < \hat{u}_1$;
- ii. $V^{AB'}(\hat{u}_1|\hat{u}_1) \geq V^{B'}(\hat{u}_1)$, and if the equality holds, $V^{AB''}(\hat{u}_1|\hat{u}_1) > V^{B''}(\hat{u}_1)$;
- iii. $V^{AB'}(u|\hat{u}_1) > V^{BAB'}(u|\hat{u}_1, u)$ for all $\hat{u}_1 < u < \hat{u}_2$;
- iv. $V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = V^{AB'}(\hat{u}_2|\hat{u}_1)$ and $V^{BAB''}(\hat{u}_2|\hat{u}_1, \hat{u}_2) > V^{AB''}(\hat{u}_2|\hat{u}_1)$;
- v. $V^{BAB'}(u|\hat{u}_1, \hat{u}_2) > \frac{1}{\phi} [\lambda_A(V_M(u + \phi/\lambda_A) - V^{BAB}(u|\hat{u}_1, \hat{u}_2)) - c]$ for all $u > \hat{u}_2$.

(b) (**Large Loss**) *If $\eta \leq \frac{1}{e-1}$, there exists $\tilde{\pi}_M(\eta)$ such that*

i. $\hat{u}_2 = \hat{u}_1 = 0$ if $\tilde{\pi}_M(\eta) \geq \pi > \frac{c}{\lambda_B}$;

ii. $\hat{u}_2 > \hat{u}_1 > 0$ if $\pi > \tilde{\pi}_M(\eta)$.

(c) **(Small Loss)** If $\frac{1}{e-1} < \eta < 1$, there exist $\tilde{\pi}_A(\eta) > \tilde{\pi}_B(\eta)$ such that

i. $\hat{u}_2 = \hat{u}_1 = 0$ if $\tilde{\pi}_B(\eta) \geq \pi > 1$;

ii. $\hat{u}_2 > \hat{u}_1 > 0$ if $\tilde{\pi}_A(\eta) \geq \pi > \tilde{\pi}_B(\eta)$;

iii. $\hat{u}_2 > \hat{u}_1 = 0$ if $\pi \geq \tilde{\pi}_B(\eta)$.

(d) The function $\mathcal{V}$ defined in (OA.1.3), with $\hat{u}_1$ and $\hat{u}_2$ derived in (a), serves as the principal's value function.

OA.1.1 Proof of Proposition 6

We begin by identifying which approach will be chosen at the deadline. Note that the execution path is chosen at the deadline if and only if $V^{B'}(0) > V^{AB'}(0|0)$. The following lemma provides the parametric condition for this.

Lemma OA.2. If $\eta \leq 1/(e-1)$, the inequality $V^{B'}(0) > V^{AB'}(0|0)$ always holds. If $\eta > 1/(e-1)$, $V^{B'}(0) > V^{AB'}(0|0)$ is equivalent to

$$\pi < \tilde{\pi}_A(\eta) \equiv \frac{e-1}{\eta(e-1)-1}.$$

Moreover, if $\pi = \tilde{\pi}_A(\eta)$, then $V^{B'}(0) = V^{AB'}(0|0)$ and $V^{B''}(0) < V^{AB''}(0|0)$.

Proof of Lemma OA.2. By $V^B(0) = V^{AB}(0|0) = 0$ and Lemma 7 (d), we have

$$\begin{aligned} \phi V^{B'}(0) &= \lambda_B \Pi - \phi - c, \\ \phi V^{AB'}(0|0) &= \lambda_A V_M\left(\frac{\phi}{\lambda_A}\right) - c = \lambda_A \left(\Pi - \frac{c}{\lambda_A} \right) (1 - e^{-1}) - \phi - c. \end{aligned}$$

Therefore, $V^{B'}(0) > V^{AB'}(0|0)$ is equivalent to:

$$(\eta(e-1) - 1) \lambda_B \Pi < c(e-1).$$

Therefore, when $\eta \leq \frac{1}{e-1}$, $V^{B'}(0) > V^{AB'}(0|0)$ always holds, and when $\eta > \frac{1}{e-1}$, $V^{B'}(0) > V^{AB'}(0|0)$ is equivalent to $\pi < \tilde{\pi}_A(\eta)$.

Next, assume that $\eta > 1/(e-1)$ and $\pi = \tilde{\pi}_A(\eta)$. With some algebra, it follows that

$$\phi^2 V^{AB''}(0|0) - \phi^2 V^{B''}(0) = \lambda_B c \left[\frac{(e-1)\eta^2}{(e-1)\eta - 1} \right].$$

The right hand side is positive from $\eta > 1/(e-1)$, thus, $V^{AB''}(0|0) > V^{B''}(0)$. $\square$

Next, we establish a condition under which the execution route is always employed. When this condition does not hold, a switch from execution to training occurs. We show the existence of the switching point $\hat{u}_1$.

Lemma OA.3. *There exists $\tilde{\pi}_M(\eta) \geq 2\lambda_B/\lambda_A = 2/(\eta+1)$ with $\tilde{\pi}_M(1) = 1$ such that the following statements hold.*

- (a) *If $1 \leq \pi < \tilde{\pi}_M(\eta)$, $V^{B'}(u) > V^{AB'}(u|u)$ for all $u \geq 0$.*
- (b) *Suppose that one of the following statements hold: (i) $\eta \leq 1/(e-1)$ and $\pi > \tilde{\pi}_M(\eta)$; (ii) $\eta > 1/(e-1)$ and $\tilde{\pi}_A(\eta) \geq \pi > \tilde{\pi}_M(\eta)$. Then, there exists $\hat{u}_1 > 0$ such that $V^{B'}(\hat{u}_1) = V^{AB'}(\hat{u}_1|\hat{u}_1)$, $V^{B''}(\hat{u}_1) < V^{AB''}(\hat{u}_1|\hat{u}_1)$ and $V^{B'}(u) > V^{AB'}(u|u)$ for all $u \in [0, \hat{u}_1)$;*

Proof of Lemma OA.3. Define $x \equiv e^{-\frac{\lambda_B}{\phi}u}$. Using (B.6) and (B.7), $\phi V^{AB'}(u|u) - \phi V^{B'}(u)$ can be expressed in the form of $H_1(x)$ with $\Delta = 1/\lambda_A$, as defined in (??) in the proof of Lemma ??. We also consider this as a function of π . With the definition of η , it can be

rewritten as follows:

$$\tilde{H}_1(x; \pi) \equiv -\{(\eta + 1)\pi - 1\} \cdot c \cdot e^{-1}x^{\eta+1} + \eta(\pi - 1) \cdot c \cdot x - (1 - \eta) \cdot c. \quad (\text{OA.1.4})$$

Observe that

$$\frac{\partial^2 \tilde{H}_1}{\partial x^2}(x; \pi) = -(\eta + 1)\eta \{(\eta + 1)\pi - 1\} \cdot c \cdot e^{-1}x^{\eta-1},$$

thus $\tilde{H}_1$ is a strict concave function in x when $\pi \geq 1$. Let $x^*(\pi)$ be the solution of $\max_x H_1(x; \pi)$ subject to $0 \leq x \leq 1$. Then, when $\pi \geq 1$, from the first order condition, we can derive that

$$x^*(\pi) = \left[\frac{\eta(\pi - 1)}{(\eta + 1)\{(\eta + 1)\pi - 1\}e^{-1}} \right]^{\frac{1}{\eta}}.^{18} \quad (\text{OA.1.5})$$

Now define

$$h(\pi) \equiv \tilde{H}_1(x^*(\pi); \pi) = K \left(\frac{\pi - 1}{(\eta + 1)\pi - 1} \right)^{\frac{1}{\eta}} (\pi - 1) \cdot c - (1 - \eta) \cdot c$$

where $K = \frac{\eta^2}{\eta + 1} \left(\frac{\eta e}{\eta + 1} \right)^{\frac{1}{\eta}}$. Observe that

$$h\left(\frac{2}{\eta + 1}\right) = (1 - \eta)c \left[\frac{\eta^2}{(\eta + 1)^2} \left(\frac{\eta(1 - \eta)e}{(\eta + 1)^2} \right)^{\frac{1}{\eta}} - 1 \right] < 0$$

from $\eta < 1$ and $\eta(1 - \eta)e \leq \frac{e}{4} < 1 \leq (\eta + 1)^2$. In addition, $\lim_{\pi \rightarrow \infty} h(\pi) = \infty$ and

$$h'(\pi) = \frac{K\pi(1 + \eta)c}{\pi - 1} \cdot \left(\frac{\pi - 1}{(\eta + 1)\pi - 1} \right)^{\frac{1}{\eta} + 1} > 0.$$

Therefore, there exists a unique π such that $h(\pi) = 0$ and $\pi \geq 2/(\eta + 1)$, and we denote this

¹⁸When $\pi \leq \frac{(\eta + 1)e^{-1} - \eta}{(\eta + 1)^2 e^{-1} - \eta}$, the solution of the maximization problem $\max_{0 \leq x \leq 1} H_1(x; \pi)$ is $x^*(\pi) = 1$. However, we can show that $\frac{(\eta + 1)e^{-1} - \eta}{(\eta + 1)^2 e^{-1} - \eta} < 1$ for any $0 < \eta$, which implies that we can focus on the interior solution when $\pi \geq 1$.

solution by $\tilde{\pi}_M(\eta)$. Also note that when $\eta = 1$, $h(2/(\eta + 1)) = h(1) = 0$ thus $\tilde{\pi}_M(1) = 1$.

- (a) Suppose that $1 \leq \pi < \tilde{\pi}_M(\eta)$. We have $0 > h(\pi) = \tilde{H}_1(x^*(\pi); \pi) \geq \tilde{H}_1(x; \pi)$ for all $0 \leq x \leq 1$. It is equivalent to $V^{B'}(u) > V^{AB'}(u|u)$ for all $u \geq 0$ in this case.
- (b) First, suppose that $\eta \leq 1/(e - 1)$ and $\pi > \tilde{\pi}_M(\eta)$. Then, we have $0 < h(\pi) = \tilde{H}_1(x^*(\pi); \pi)$. In addition, by Lemma OA.2, we have $\tilde{H}_1(1; \pi) = \phi(V^{AB'}(0|0) - V^{B'}(0)) < 0$. Then, by concavity of $\tilde{H}_1$ w.r.t. x and $\frac{\partial \tilde{H}_1}{\partial x}(x^*(\pi); \pi) = 0$, there exists $x_1 \in (x^*(\pi), 1]$ such that $\tilde{H}_1(x_1; \pi) = 0$, $\frac{\partial \tilde{H}_1}{\partial x}(x_1; \pi) < 0$ and $\tilde{H}_1(x; \pi) < 0$ for all $x \in (x_1, 1]$. By defining $\hat{u}_1 \equiv -(\phi/\lambda_B) \log x_1$, the above conditions can be translated into: $V^{B'}(\hat{u}_1) = V^{AB'}(\hat{u}_1|\hat{u}_1)$, $V^{B''}(\hat{u}_1) < V^{AB''}(\hat{u}_1|\hat{u}_1)$ and $V^{B'}(u) > V^{AB'}(u|u)$ for all $u \in [0, \hat{u}_1)$.

Next, suppose that $\eta > 1/(e - 1)$. Note that by the definition of $\tilde{\pi}_A(\eta)$, if $\pi \geq \tilde{\pi}_A(\eta)$, $\tilde{H}_1(1; \pi) \geq 0$. It implies that $h(\pi) \geq \tilde{H}_1(1; \pi) \geq 0$ and $\pi \geq \tilde{\pi}_M(\eta)$. Therefore, we can see that $\tilde{\pi}_A(\eta) \geq \tilde{\pi}_M(\eta)$. If $\tilde{\pi}_A(\eta) \geq \pi > \tilde{\pi}_M(\eta)$, we also have $\tilde{H}_1(x^*(\pi); \pi) > 0 > \tilde{H}_1(1; \pi)$. By using the same arguments as above, we can show that there exists $\hat{u}_1 > 0$ such that $V^{B'}(\hat{u}_1) = V^{AB'}(\hat{u}_1|\hat{u}_1)$, $V^{B''}(\hat{u}_1) < V^{AB''}(\hat{u}_1|\hat{u}_1)$ and $V^{B'}(u) > V^{AB'}(u|u)$ for all $u \in [0, \hat{u}_1)$.

□

The following lemma shows that when there is an efficiency loss from training and the training path is employed, there will be an additional switching point, $\hat{u}_2$.

Lemma OA.4. *Suppose that $\eta < 1$, $\Pi > c/\lambda_B$ and one of the followings hold: (i) $V^{B'}(\hat{u}_1) < V^{AB'}(\hat{u}_1|\hat{u}_1)$; (ii) $V^{B'}(\hat{u}_1) = V^{AB'}(\hat{u}_1|\hat{u}_1)$ and $V^{B''}(\hat{u}_1) < V^{AB''}(\hat{u}_1|\hat{u}_1)$. Then, there exists $\hat{u}_2 > \hat{u}_1$ such that $V^{AB'}(\hat{u}_2|\hat{u}_1) = V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and $V^{AB''}(\hat{u}_2|\hat{u}_1) < V^{BAB''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and such $\hat{u}_2$ is unique. Moreover, $V^{AB'}(u|\hat{u}_1) > V^{BAB'}(u|\hat{u}_1, u)$ for all $u \in (\hat{u}_1, \hat{u}_2)$.*

Proof of Lemma OA.4. Using (OA.1.2), (B.7) and $V^{BAB}(u|\hat{u}_1, u) = V^{AB}(u|\hat{u}_1)$, $\phi V^{BAB'}(u|\hat{u}_1, u) - \phi V^{AB'}(u|\hat{u}_1)$ can be rewritten as follows:

$$\lambda_B \Pi - \lambda_A \{V_M(u + \phi/\lambda_A) + u + \phi/\lambda_A\} + (\lambda_A - \lambda_B)(V^{AB}(u|\hat{u}_1) + u).$$

By performing a similar derivation as in (??) and using $\eta = \frac{\lambda_A}{\lambda_B} - 1$ and $y \equiv e^{-\frac{\lambda_A}{\phi}(u - \hat{u}_1)}$, the above expression can be further rewritten as follows:

$$\begin{aligned} \tilde{H}_2(y) \equiv & \frac{1 - \eta}{1 + \eta} c + (\lambda_A \Pi - c) e^{-1 - \frac{\lambda_A}{\phi} \hat{u}_1} \left[1 + \frac{\eta}{1 + \eta} \log y \right] y \\ & + \eta \left[\frac{1 - \eta}{1 + \eta} c - (\lambda_B \Pi - c) e^{-\frac{\lambda_B}{\phi} \hat{u}_1} \right] y. \end{aligned} \tag{OA.1.6}$$

Note that $\tilde{H}_2(1) = \phi V^{BAB'}(\hat{u}_1|\hat{u}_1, \hat{u}_1) - \phi V^{AB'}(\hat{u}_1|\hat{u}_1) = \phi V^{B'}(\hat{u}_1) - \phi V^{AB'}(\hat{u}_1|\hat{u}_1) \leq 0$ by assumption. By differentiating $\tilde{H}_2$ twice, we have

$$\tilde{H}_2''(y) = \frac{\eta}{1 + \eta} (\lambda_A \Pi - c) e^{-1 - \frac{\lambda_A}{\phi} \hat{u}_1} \frac{1}{y} > 0.$$

Since $\Pi > \frac{c}{\lambda_B} > \frac{c}{\lambda_A}$, $\tilde{H}_2$ is strictly convex in y . Also note that

$$\lim_{y \rightarrow 0} \tilde{H}_2(y) = \frac{1 - \eta}{1 + \eta} c > 0.$$

By the convexity of $\tilde{H}_2$, there exists $y_2 \in (0, 1)$ such that (i) $\tilde{H}_2(y) < 0$ for all $y \in (y_2, 1)$, (ii) $\tilde{H}_2(y_2) = 0$, and (iii) $\tilde{H}_2'(y_2) < 0$. Let $\hat{u}_2 = \hat{u}_1 - \frac{\phi}{\lambda_A} \log y_2$. Then, from (i) and (ii), we have $V^{AB'}(u|\hat{u}_1) > V^{BAB'}(u|\hat{u}_1, u)$ for all $u \in (\hat{u}_1, \hat{u}_2)$ and $V^{AB'}(\hat{u}_2|\hat{u}_1) = V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. Additionally, since y is decreasing in u , $\tilde{H}_2'(y_2) < 0$ implies that $V^{AB''}(\hat{u}_2|\hat{u}_1) < V^{BAB''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. □

Next, when there is a switching point $\hat{u}_2$, the following lemma shows that the execution path is employed for all $u > \hat{u}_2$.

Lemma OA.5. Suppose that $\pi > 1$, $V^{AB'}(\hat{u}_2|\hat{u}_1) = V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and $V^{AB''}(\hat{u}_2|\hat{u}_1) < V^{BAB''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. Then, $\lambda_A (V_M(u + \phi/\lambda_A) - V^{BAB}(u|\hat{u}_1, \hat{u}_2)) - \phi V^{BAB'}(u|\hat{u}_1, \hat{u}_2) - c < 0$ for all $u > \hat{u}_2$.

Proof of Lemma OA.5. By differentiating (B.6) and (B.7), we have

$$\begin{aligned}\phi V^{AB''}(u|\hat{u}_1) &= \lambda_A (V'_M(u + \phi/\lambda_A) + 1) - \lambda_A (V^{AB'}(u|\hat{u}_1) + 1), \\ \phi V^{BAB''}(u|\hat{u}_1, \hat{u}_2) &= -\lambda_B (V^{BAB'}(u|\hat{u}_1, \hat{u}_2) + 1).\end{aligned}$$

Then, $V^{AB'}(\hat{u}_2|\hat{u}_1) = V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and $V^{AB''}(\hat{u}_2|\hat{u}_1) < V^{BAB''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ imply that

$$\begin{aligned}(\lambda_A - \lambda_B)(1 + V^{AB'}(\hat{u}_2|\hat{u}_1)) &> \lambda_A(V'_M(\hat{u}_2 + \phi/\lambda_A) + 1) \\ \iff \eta(1 + V^{AB'}(\hat{u}_2|\hat{u}_1)) &> (\eta + 1) \left(\frac{\lambda_A \Pi - c}{\phi} \right) e^{-\frac{\lambda_A}{\phi} \hat{u}_2 - 1}. \quad (\text{OA.1.7})\end{aligned}$$

Define a function $H_3 : [\hat{u}_2, \infty) \rightarrow \mathbb{R}$ as

$$H_3(u) \equiv \lambda_A [V_M(u + \phi/\lambda_A) - V^{BAB}(u|\hat{u}_1, \hat{u}_2)] - \phi V^{BAB'}(u|\hat{u}_1, \hat{u}_2) - c.$$

With some algebra, we can derive that

$$H_3(u) = (\eta - 1)c - (\lambda_A \Pi - c)e^{-\frac{\lambda_A}{\phi} \hat{u}_2 - 1} \cdot e^{\frac{\lambda_A}{\phi}(\hat{u}_2 - u)} + \eta \phi (V^{AB'}(\hat{u}_2|\hat{u}_1) + 1) e^{\frac{\lambda_B}{\phi}(\hat{u}_2 - u)}.$$

Also note that

$$\begin{aligned}H_3(\hat{u}_2) &= \lambda_A [V_M(\hat{u}_2 + \phi/\lambda_A) - V^{AB}(\hat{u}_2|\hat{u}_1)] - c - \phi V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2) \\ &= \phi V^{AB'}(\hat{u}_2|\hat{u}_1) - \phi V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = 0.\end{aligned}$$

Define $x \equiv e^{\frac{\lambda_B}{\phi}(\hat{u}_2 - u)}$. Then, $H_3(u)$ can be rewritten as follows:

$$\tilde{H}_3(x) = (\eta - 1)c - (\lambda_A \Pi - c)e^{-\frac{\lambda_A}{\phi}\hat{u}_2 - 1}x^{\eta+1} + \eta\phi \left(V^{AB'}(\hat{u}_2|\hat{u}_1) + 1 \right) x$$

and $\tilde{H}_3(1) = H_3(\hat{u}_2) = 0$.

Note that

$$\tilde{H}_3'(x) = -(\eta + 1)(\lambda_A \Pi - c)e^{-\frac{\lambda_A}{\phi}\hat{u}_2 - 1}x^\eta + \eta\phi \left(V^{AB'}(\hat{u}_2|\hat{u}_1) + 1 \right).$$

By (OA.1.7), we can derive that

$$\tilde{H}_3'(1) = -(\eta + 1)(\lambda_A \Pi - c)e^{-\frac{\lambda_A}{\phi}\hat{u}_2 - 1} + \eta\phi \left(V^{AB'}(\hat{u}_2|\hat{u}_1) + 1 \right) > 0.$$

Also note that

$$\tilde{H}_3''(x) = -(\eta + 1)\eta(\lambda_A \Pi - c)e^{-\frac{\lambda_A}{\phi}\hat{u}_2 - 1}x^{\eta-1} < 0.$$

Therefore, $\tilde{H}_3'(x) > 0$ for all $0 < x < 1$. Since $\tilde{H}_3(1) = 0$, $\tilde{H}_3(x) < 0$ for all $x \in (0, 1)$. Thus, $\lambda_A (V_M(u + \phi/\lambda_A) - V^{BAB}(u|\hat{u}_1, \hat{u}_2)) - \phi V^{BAB'}(u|\hat{u}_1, \hat{u}_2) - c < 0$ for all $u \geq \hat{u}_2$. $\square$

Lastly, we show that the resulting value function is concave, and that L^B and L^A —the functions specifying deviations, defined in (B.13) and (B.14)—are nonpositive.

Lemma OA.6. *Suppose that $\pi > 1$ and $\eta < 1$.*

(a) $\mathcal{V}$ is concave;

(b) for any $u \geq 0$, $L^B(u, R) \leq 0$ for all $R \geq u + \phi/\lambda_B$, and $L^A(u, w) \leq 0$ for all $w \geq u + \phi/\lambda_A$.

Proof of Lemma OA.6. (a) When $u < \hat{u}_1$, $\mathcal{V}''(u) = V^{B''}(u) < 0$ from Lemma 8 (c).

When $\hat{u}_1 < u < \hat{u}_2$, the inequality (B.12) is still applicable, and from $\lambda_A < 2\lambda_B$, we have $\mathcal{V}''(u) = V^{AB''}(u|\hat{u}_1) \leq 0$.

When $u > \hat{u}_2$, by differentiating (OA.1.1) twice, we have

$$V^{BAB''}(u|\hat{u}_1, \hat{u}_2) = - \left(\frac{\lambda_B}{\phi} \right)^2 \cdot \left(\Pi - \frac{c}{\lambda_B} - (V^{AB}(\hat{u}_2|\hat{u}_1, \hat{u}_2) + \hat{u}_2) \right) e^{-\frac{\lambda_B}{\phi}(u-\hat{u}_2)}.$$

Note that $V^{AB}(\hat{u}_2|\hat{u}_1, \hat{u}_2) + \hat{u}_2$ cannot exceed the first-best expected surplus $\Pi - c/\lambda_B$, thus, the above expression is negative.

Since these component functions are smoothly pasted at $\hat{u}_1$ and $\hat{u}_2$, the entire value function is concave.

- (b) As in the no efficiency loss case, $\mathcal{V}(u) + u$ is increasing in u , thus, $\mathcal{V}'(u) \geq -1$ and it gives $\frac{\partial L^B}{\partial R} \leq 0$. Thus, it is sufficient to show that $L^B(u, u + \phi/\lambda_B) \leq 0$ for all $u \geq 0$.

Observe that from (B.6), (B.7) and (OA.1.2), we have

$$L^B(u, u + \phi/\lambda_B) = \begin{cases} 0, & \text{if } u \leq \hat{u}_1 \text{ or } u \geq \hat{u}_2, \\ \phi V^{BAB'}(u|\hat{u}_1, u) - \phi V^{AB'}(u|\hat{u}_1), & \text{if } u \in (\hat{u}_1, \hat{u}_2). \end{cases}$$

Since $\hat{u}_1$ and $\hat{u}_2$ are chosen to satisfy $V^{BAB'}(u|\hat{u}_1, u) < V^{AB'}(u|\hat{u}_1)$ for all $u \in (\hat{u}_1, \hat{u}_2)$, L^B is always nonpositive.

Likewise, from the concavity of $\mathcal{V}$, $\frac{\partial L^A}{\partial u} \geq 0$. Thus, it is sufficient to show that $L^A(u, u + \phi/\lambda_A) \leq 0$ for all $u \geq 0$. Then, we have

$$L^A(u, u + \phi/\lambda_A) = \begin{cases} \phi V^{AB'}(u|u) - \phi V^{B'}(u), & \text{if } u \leq \hat{u}_1, \\ 0, & \text{if } u \in (\hat{u}_1, \hat{u}_2), \\ \lambda_A(V_M(u + \phi/\lambda_A) - V^{BAB}(u|\hat{u}_1, \hat{u}_2)) - c - \phi V^{BAB'}(u|\hat{u}_1, \hat{u}_2), & \text{if } u \geq \hat{u}_2. \end{cases}$$

By Lemma OA.3 and Lemma OA.5, L^A is always nonpositive.

□

Now we prove Proposition 6.

Proof of Proposition 6. We start by showing that, for each condition in (b) and (c), the switching points are as stated and the conditions in (a) also hold.

(b-i & c-i) By Lemma OA.3 (a), $V^{B'}(u) > V^{AB'}(u|u)$ for all $u > 0$. Note that $V^B(u) = V^{BAB}(u|0, 0)$ and

$$\phi V^{AB'}(u|u) = \lambda_A(V_M(u + \phi/\lambda_A) - V^{BAB}(u|0, 0)) - c$$

by (B.7) and $V^{AB}(u|u) = V^B(u)$. Therefore, with $\hat{u}_1 = \hat{u}_2 = 0$, the conditions (a-i)–(a-iv) hold trivially, and the condition (a-v) holds as demonstrated above.

(b-ii & c-ii) By Lemma OA.3 (b), there exists $\hat{u}_1 > 0$ such that the conditions (a-i) and (a-ii) hold. Next, by Lemma OA.4, there exists $\hat{u}_2 > \hat{u}_1$ such that the conditions (a-iii) and (a-iv) hold. By Lemma OA.5, the condition (a-v) holds.

(c-iii) By Lemma OA.2, $V^{AB'}(0|0) > V^B(0)$. By setting $\hat{u}_1 = 0$, the conditions (a-i) and (a-ii) hold trivially. Next, by Lemma OA.4, there exists $\hat{u}_2 > 0$ such that the conditions (a-iii) and (a-iv) hold. By Lemma OA.5, the condition (a-v) holds.

(d) Following the same steps of the proof of Proposition 2, $\mathcal{V}$ solves (HJB $\mathcal{V}$) subject to (IC). □

OA.2 Proofs of Proposition 4 and Proposition 5

Lemma OA.7. *Suppose that $\pi > \tilde{\pi}_M(\eta)$ and $\eta \leq c/(c + \phi)$. Then, $\hat{u}_2$ is less than $\bar{u}$.*

Proof of Lemma OA.7. Since $\mathcal{V}$ is strictly concave, $\hat{u}_2 < \bar{u}$ is equivalent to $0 < \mathcal{V}'(\hat{u}_2) = V^{AB'}(\hat{u}_2|\hat{u}_1) = V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. Then, $0 < V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ is equivalent to:

$$\lambda_B(\hat{u}_2 + V^{AB}(\hat{u}_2|\hat{u}_1)) < \lambda_B\Pi - c - \phi. \quad (\text{OA.2.1})$$

Also note that $V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = V^{BAB'}(\hat{u}_2|\hat{u}_1)$ and $V^{BAB}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = V^{AB}(\hat{u}_2|\hat{u}_1)$ imply that

$$\lambda_B(\Pi - \hat{u}_2 - V^{AB}(\hat{u}_2|\hat{u}_1)) = \lambda_A(V_M(\hat{u}_2 + \phi/\lambda_A) + \hat{u}_2 + \phi/\lambda_A) - \lambda_A(V^{AB}(\hat{u}_2|\hat{u}_1) + \hat{u}_2)$$

by (B.6) and (B.7). By plugging (B.1) into the above equation, we can derive that

$$\begin{aligned} (\lambda_A - \lambda_B)(V^{AB}(\hat{u}_2|\hat{u}_1) + \hat{u}_2) &= \lambda_A \left(\Pi - \frac{c}{\lambda_A} \right) \left(1 - e^{-\frac{\lambda_A}{\phi}\hat{u}_2-1} \right) - \lambda_B\Pi \\ \iff \eta\lambda_B(V^{AB}(\hat{u}_2|\hat{u}_1) + \hat{u}_2) &= \eta\lambda_B\Pi - c - (\lambda_A\Pi - c)e^{-\frac{\lambda_A}{\phi}\hat{u}_2-1}. \end{aligned}$$

Then, by plugging this into (OA.2.1), $0 < V^{BAB'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ is equivalent to

$$\eta(c + \phi) - c < (\lambda_A\Pi - c)e^{-\frac{\lambda_A}{\phi}\hat{u}_2-1}.$$

Since $\Pi > c/\lambda_B > c/\lambda_A$, the right hand side of the above inequality is always greater than 0. Since it is assumed that $\eta \leq c/(c + \phi)$, the left hand side of the above inequality is always less than or equal to 0. Therefore, the above inequality always holds, i.e., $\hat{u}_2$ is less than $\bar{u}$. $\square$

Lemma OA.8. Suppose that $\eta > \bar{\eta} = \max\{1/(e-1), \sqrt{c/(c+\phi)}\}$. There exists $\tilde{\pi}_B(\eta) \in (\tilde{\pi}_M(\eta), \tilde{\pi}_A(\eta))$ such that $\hat{u}_1 < \bar{u}$ if and only if $\pi > \tilde{\pi}_B(\eta)$.

Proof of Lemma OA.8. From $\eta > 1/(e-1)$, $\tilde{\pi}_A(\eta)$ exists. Suppose that $\pi \geq \tilde{\pi}_A(\eta)$. By Lemma OA.2, $\hat{u}_1 = 0$. Note that

$$\tilde{\pi}_A(\eta) = \frac{e-1}{(e-1)\eta-1} \geq \frac{e-1}{e-2} > 2 \geq \frac{c+\phi}{c} = \pi_F.$$

Then, the project is feasible and $\bar{u}$ is greater than 0, i.e., $\bar{u} > \hat{u}_1$.

Now suppose that $\pi_M(\eta) < \pi < \pi_A(\eta)$. Since V is strictly concave, $\hat{u}_1 < \bar{u}$ is equivalent to $0 < \mathcal{V}'(\hat{u}_1) = V^{B'}(\hat{u}_1)$. Note that $0 < V^{B'}(\hat{u}_1)$ is equivalent to:

$$\frac{\phi}{\lambda_B \Pi - c} < e^{-\frac{\lambda_B}{\phi} \hat{u}_1} = \hat{x}_1. \quad (\text{OA.2.2})$$

Recall that $\hat{x}_1$ is a solution where $\tilde{H}_1(x)$, as defined in (OA.1.4), equals zero. Additionally, $\pi_M(\eta) < \pi < \pi_A(\eta)$ implies that $\tilde{H}_1(1) < 0 < \tilde{H}_1(x^*)$ where x^* is defined in (OA.1.5).¹⁹

There are two possible cases that satisfy (OA.2.2): (i) $x^* \geq \frac{\phi}{\lambda_B \Pi - c}$; (ii) $\frac{\phi}{\lambda_B \Pi - c} > x^*$ and $\tilde{H}_1(\frac{\phi}{\lambda_B \Pi - c}) < 0$.

The first case is equivalent to $\tilde{H}'_1(\frac{\phi}{\lambda_B \Pi - c}) < 0$. By algebra, we can show that it is equivalent to

$$\frac{(\eta+1)\pi-1}{(\pi-1)^{\eta+1}} < \frac{\eta e}{\eta+1} \phi^{-\eta} c^\eta. \quad (\text{OA.2.3})$$

The second case is equivalent to $\tilde{H}'_1(\frac{\phi}{\lambda_B \Pi - c}) \geq 0$ and $\tilde{H}_1(\frac{\phi}{\lambda_B \Pi - c}) < 0$. By algebra, we can show that it is equivalent to

$$\frac{\eta e}{\eta+1} \phi^{-\eta} c^\eta \leq \frac{(\eta+1)\pi-1}{(\pi-1)^{\eta+1}} < (\eta(c+\phi)-c) e \phi^{-\eta-1} c^\eta. \quad (\text{OA.2.4})$$

¹⁹For simplicity, Π is omitted from the definition of $\tilde{H}_1$ and x^* .

Last, by the proof of Lemma OA.3, we can show that $\pi > \pi_M(\eta)$ is equivalent to

$$\frac{(\eta + 1)\pi - 1}{(\pi - 1)^{\eta+1}} < \left(\frac{\eta^2}{1 - \eta^2} \right)^\eta \frac{\eta e}{1 + \eta}. \quad (\text{OA.2.5})$$

Now we compare the above three conditions. Using $\eta > \sqrt{c/(c + \phi)}$, by simple algebra, we can show that

$$\frac{\eta e}{\eta + 1} \phi^{-\eta} c^\eta < (\eta(c + \phi) - c) e \phi^{-\eta-1} c^\eta < \left(\frac{\eta^2}{1 - \eta^2} \right)^\eta \frac{\eta e}{1 + \eta}.$$

Therefore, the inequality

$$\frac{(\eta + 1)\pi - 1}{(\pi - 1)^{\eta+1}} < (\eta(c + \phi) - c) e \phi^{-\eta-1} c^\eta$$

imply that (OA.2.3), (OA.2.4) and (OA.2.5). Define $\pi_B(\eta)$ be the value of π that makes both sides of the above inequality equal. Then, $\pi_B(\eta) > \pi_M(\eta)$ since $\pi < \pi_M(\eta)$ implies $\pi < \pi_B(\eta)$. Therefore, there exists $\pi_B(\eta) > \pi_M(\eta)$ such that $\hat{u}_1 < \bar{u}$ if and only if $\pi > \pi_B(\eta)$. $\square$

Lemma OA.9. *Suppose that $\eta > \bar{\eta} = \sqrt{c/(c + \phi)}$. Then, $\hat{u}_2 \geq \bar{u}$.*

Proof of Lemma OA.9. By following the proof of Lemma OA.7, $\hat{u}_2 \geq \bar{u}$ is equivalent to

$$\bar{y} \equiv \frac{(\eta - 1)c + \eta\phi}{(\lambda_A \Pi - c)e^{-\frac{\lambda_A}{\phi} \hat{u}_1 - 1}} \geq e^{\frac{\lambda_A}{\phi} (\hat{u}_1 - \hat{u}_2)} = \hat{y}_2 \quad (\text{OA.2.6})$$

By the proof of Lemma OA.4, $\hat{y}_2$ is the solution, which is not equal to 1, of $\tilde{H}_2(y) = 0$.²⁰ Since $\hat{u}_2 \geq \hat{u}_1$, if $\bar{y} \geq 1$, the above inequality holds, thus, we restrict attention to the case of $\bar{y} < 1$. Observe that the inequality $\tilde{H}_2(\bar{y}) \leq 0$ implies (OA.2.6) because $\tilde{H}_2$ is strictly convex in y and $\tilde{H}_2(1) \leq 0$.

²⁰The function $\tilde{H}_2$ is defined in (OA.1.6).

Using the definition of $\tilde{H}_1$ in (OA.1.4) and $\hat{x}_1 \equiv e^{-\frac{\lambda_B}{\phi} \hat{u}_1}$, $\tilde{H}_2(y)$ can be rewritten as follows:

$$\tilde{H}_2(y) = \frac{1-\eta}{1+\eta}c - \tilde{H}_1(\hat{x}_1)y + \left[-\frac{1-\eta}{1+\eta}c + \frac{\eta}{1+\eta}(\lambda_A\Pi - c)e^{-\frac{\lambda_A}{\phi}\hat{u}_1-1} \log y \right] y.$$

Also note that $\hat{x}_1$ is chosen to satisfy $\tilde{H}_1(\hat{x}_1)$ being greater than equal to zero.

By plugging the definition of $\bar{y}$ into the above equation, we can derive that

$$\tilde{H}_2(\bar{y}) = \frac{1-\eta}{1+\eta}c(1-\bar{y}) - \tilde{H}_1(\hat{x}_1)\bar{y} + \frac{\eta}{1+\eta}((\eta-1)c + \eta\phi) \log \bar{y}.$$

Now define a new function G as follows:

$$G(y) \equiv \frac{1-\eta}{1+\eta}c(1-y) - \tilde{H}_1(\hat{x}_1)y + \frac{\eta}{1+\eta}((\eta-1)c + \eta\phi) \log y,$$

and it is enough to show that $G(y) \leq 0$ for all $y < 1$.

Note that

$$G''(y) = -\frac{\eta}{1+\eta} \left(\frac{(\eta-1)c + \eta\phi}{y^2} \right) < 0$$

from $\eta \geq \sqrt{c/(c+\phi)} > c/(c+\phi)$. Also note that

$$G'(1) = -\tilde{H}_1(\hat{x}_1) + \frac{1}{1+\eta}((\eta^2-1)c + \eta^2\phi) < 0.$$

from $\eta \geq \sqrt{c/(c+\phi)}$ and $\tilde{H}_1(\hat{x}_1) \geq 0$. Lastly, note that $G(1) = -\tilde{H}_1(\hat{x}_1) \leq 0$. Therefore, for all $y < 1$, $G(y) \leq G(1) + G'(1)(1-y) \leq 0$. Therefore, $\tilde{H}_2(\bar{y}) \leq 0$ and $u_2 \geq \bar{u}$. $\square$

Now we prove Proposition 4 and Proposition 5.

Proof of Proposition 4. Note that $\hat{u}_2$ is always greater than $\bar{u}$ by Lemma OA.9 since $\eta >$

$\sqrt{c/(c+\phi)}$. Additionally, using Lemma OA.2, Lemma OA.3 and Lemma OA.8, we have

$$\mathcal{V}(\bar{u}) = \begin{cases} V^B(\bar{u}), & \text{if } \pi_F < \pi < \tilde{\pi}_B(\eta), \\ V^{AB}(\bar{u}|\hat{u}_1), & \text{if } \tilde{\pi}_B(\eta) < \pi < \tilde{\pi}_A(\eta), \\ V^{AB}(\bar{u}|0), & \text{if } \tilde{\pi}_A(\eta) < \pi. \end{cases}$$

As in Theorem 1, the value functions above can be implemented by execution-only, one-switch and training-only contracts, respectively. $\square$

Proof of Proposition 5. By Proposition 6 (b-i), when $\pi \leq \tilde{\pi}_M(\eta)$, $\hat{u}_1 = \hat{u}_2 = 0$. By Proposition 6 (b-ii) and Lemma OA.7, when $\pi > \tilde{\pi}_M(\eta)$, $\bar{u} > \hat{u}_2 > \hat{u}_1 > 0$. Therefore,

$$\mathcal{V}(\bar{u}) = \begin{cases} V^{BAB}(\bar{u}|0,0) = V^B(\bar{u}), & \text{if } \pi_F < \pi \leq \tilde{\pi}_M(\eta), \\ V^{BAB}(\bar{u}|\hat{u}_1, \hat{u}_2), & \text{if } \tilde{\pi}_M(\eta) < \pi. \end{cases}$$

By Lemma OA.1, $(V^{BAB}(\bar{u}|\hat{u}_1, \hat{u}_2), \bar{u})$ can be implemented by a two-switch contract. Therefore, when $\pi \in (\pi_F, \tilde{\pi}_M(\eta)]$, the execution-only contract is optimal, and when $\pi > \tilde{\pi}_M(\eta)$, there exists a two-switch contract that is optimal. $\square$